Scanine: A Mobile-Integrated Application for Multi-Class Canine Dermatological Screening via Grad-CAM and Condition-Adaptive First-Response Guidance

---

A Capstone Project Presented to the Faculty
of the IT and Computer Education Unit
Leyte Normal University
Tacloban City

In Partial Fulfillment of the
Requirement for the Degree
Bachelor of Science in Information Technology

---

by:

Chloe Ivonne S. Verterra I
Nicole I. Severino
Blecy M. Copioso
Chielsy Jul Java

May 2026

-

Table of Contents

Title Page
Table of Contents **2**
List of Figures **4**
List of Tables **5**
CHAPTER 1 **1**
Introduction 1
1.2 Purpose and Description 4
1.3 Objective of the Study 6
1.4 Scope and Limitation 7
CHAPTER 2 **9**
Review of Related Literature **9**
2.1 Clinical Profile of Target Canine Dermatological Conditions 9
2.2 Smartphone-Based Image Acquisition and Quality Assessment for Veterinary Screening 11
2.3 Lightweight Deep Learning Architectures and Attention Mechanisms for Mobile Veterinary AI 12
2.4 Explainable Artificial Intelligence through Gradient-Weighted Class Activation Mapping (Grad-CAM) 14
2.5 Native Mobile Edge AI and On-Device Inference Frameworks 15
Review of Related Systems **17**
2.6 Multi-Class Canine Dermatological Screening Systems 17
2.7 Mobile Deep Learning Classification Systems with Attention Mechanisms 18
2.8 Explainable AI-Based Veterinary Diagnostic Support Systems 19
2.9 Native Mobile Edge AI Screening Systems 20
2.10 AI-Assisted First-Response and Community Veterinary Support Systems 21
CHAPTER 3 **22**
3.1 Current System 22
3.2 Proposed System 24
3.2.1 Software 25
3.2.2 Hardware 27
3.2.3 Peopleware 28
CHAPTER 4 **29**
4.1 Requirements Analysis **29**
4.1.1 Performance Requirements 29
4.1.2 Safety Requirements 31
4.1.3 Security Requirements 33
4.2 Design of Software, Systems, Product, and/or Processes **34**
Systems Development Life Cycle 35

Requirements Analysis 37
Activity List 37
Gantt Chart 42
Context Diagram 46
Data Flow Diagrams 49
Image Processing Pipeline 57
Participant Selection 66
Ethical Considerations 66
Use Case 68
Implementation Plan 69
REFERENCES **76**

List of Figures

Figure 4.1. SDLC 36
Figure 4.2. PERT Chart 45
Figure 4.3. Proposed Context Diagram 47
Figure 4.4. Manual Process Context Diagram 48
Figure 4.5. Level 1 Data Flow Diagram (DFD) 50
Figure 4.6. Level 2 DFD: Process 1 52
Figure 4.7. Level 2 DFD: Process 2 54
Figure 4.8. Level 2 DFD: Process 3 55
Figure 4.9. Level 2 DFD: Process 4 56
Figure 4.10. Image Processing Pipeline 63
Figure 4.11. Use-Case 69

List of Tables

Table 3.1. Software 25
Table 3.2. Hardware 27
Table 4.1. Performance Requirements of the Proposed System 29
Table 4.2. Safety Requirements 31
Table 4.3. Security Requirements 33
Table 4.4. Activity List 38
Table 4.5. Gantt Chart 44
Table 4.6. 5-Point Likert Scale 65
Table 4.7. Participant Inclusion and Exclusion Criteria 66
Table 4.8. Implementation Plan (Tabular) 72

CHAPTER 1

This chapter presents the zoonotic and diagnostic context surrounding canine dermatophytosis, hotspots, and flea allergy dermatitis, identifies the gaps in existing automated multi-class screening approaches, and outlines the technical and clinical foundations underlying the development of Scanine as a field-deployable, explainability-driven triage tool with condition-adaptive first-response guidance for community responders.

Introduction

Canine skin diseases have a way of going unnoticed until the damage is already done, and among dogs worldwide, three conditions in particular, dermatophytosis, acute moist dermatitis (commonly known as hotspot), and flea allergy dermatitis, tend to progress well past their earliest, most manageable stages before anyone intervenes (Paryuni et al., 2020; Cho, 2025; Sri et al., 2025). Left unchecked, this delay compounds the very problem that early screening is meant to catch, allowing a condition to advance or spread through a population in the time it takes to be noticed. However, the diagnostic methods most relied upon to confirm any of these conditions, fungal culture, microscopy, and direct veterinary examination, are not designed for that earliest point of contact; they depend on laboratory access and specialist interpretation that community responders and animal welfare personnel rarely have on hand (Paryuni et al., 2020; Pal, 2025). Accordingly, this study proceeds from the premise that the gap in canine dermatological screening is not primarily diagnostic but one of deployment, and that closing it calls for a non-invasive, community-accessible framework capable of producing interpretable, condition-specific results at the point of first animal contact.

This deployment gap is not confined to any single setting. Stray and community dog populations have been documented internationally as reservoirs for these and other zoonotic conditions, particularly in urban environments where human-animal contact is close and frequent (Khaliq et al., 2026). This pattern points to something structural rather than an isolated shortfall in any one region: the tools capable of confirming these conditions were built around laboratory workflows, not around the point where dogs and their handlers actually meet. Wherever that point of contact happens, the same shortfall applies, and no non-invasive, field-ready alternative has taken its place.

Automated image-based classification has emerged as a plausible alternative to diagnostic methods that field settings cannot support. Convolutional neural network systems applied to canine skin images have reported classification accuracies between 83% and 98.9%, outperforming manual assessment in speed and consistency across a range of architectures (Serradj, 2025; Sri et al., 2025), and lightweight models trained on smartphone-captured images have shown that reasonably reliable outputs are achievable without laboratory infrastructure or a persistent network connection (Marquardt et al., 2026). Whether a smartphone can classify a canine skin condition is largely settled at this point. What remains unresolved is whether that classification gives a non-expert anything they can actually use. Existing systems tend to reduce dermatophytosis, acute moist dermatitis, and flea allergy dermatitis to interchangeable output labels, assigning a category without surfacing the visual cues that separate one condition from another. The moist, rapidly spreading borders of an acute moist dermatitis lesion look different from the dry, circular, scaling pattern of a dermatophyte infection, which in turn differs from the dorsal, papular distribution typical of flea

allergy dermatitis. These differences matter in the field: they inform how urgently a responder needs to act and what to do while waiting for a veterinarian. In much of this work, accuracy has been the benchmark, and interpretive value has been treated as secondary.

Gradient-weighted Class Activation Mapping has been introduced into this body of work as a way of making model decisions visible. By producing spatial heatmaps that highlight the image regions most responsible for a model's classification, Grad-CAM offers a responder something more concrete than a bare label: a visual indication of where the pathological activity appears to be on the animal (Garcia et al., 2024; Badhon et al., 2025). The technique has been validated in lightweight, mobile-integrated deployments, where heatmaps directed non-expert attention toward lesion-consistent regions during real-time classification (Revathy et al., 2025). Even so, a responder who can see roughly where a lesion sits still does not know what to do about it, and the window between first contact and professional veterinary consultation is where condition-specific guidance matters most. A dog suspected of ringworm needs to be isolated to limit zoonotic spread; a dog with flea allergy dermatitis needs environmental flea control; a dog with a hotspot needs to be kept from further self-trauma that would worsen the wound. Few tools pair this kind of guidance with content reviewed for clinical safety; those that attempt some form of it, such as Dissanayaka et al. (2022), rely on general home remedy suggestions rather than veterinarian-reviewed, condition-matched care. This absence carries particular weight locally: the Department of Health formally lists dermatophytosis among the country's primary zoonotic concerns alongside rabies, yet responders in areas like Tacloban still have no standardized, non-invasive way to act on that risk at the point of

first contact (DOH, 2026). What the literature offers, in effect, are separate pieces: classification models, visualization layers, and clinical references that no one screening an animal in the field would realistically stop to consult. No system has assembled these into one coherent triage output suited to the conditions under which community-level screening actually happens.

Taken together, these are not isolated shortcomings so much as a mismatch between what existing systems were built to do and what community-level canine dermatological screening actually requires. A system that classifies without distinguishing, detects without explaining, and visualizes without guiding falls short of the problem it is meant to address. However, closing this gap calls for a system built around what a field responder needs at the point of first contact, rather than an adjustment to what already exists. In response, this study introduces Scanine, a mobile application built for multi-class canine dermatological screening among community responders in Tacloban City.

1.2 Purpose and Description

The primary purpose of this study is to develop Scanine, an offline mobile application designed to assist community responders, including pet owners, animal welfare volunteers, and community handlers, in the preliminary screening of three common canine dermatological conditions through offline multi-class image classification. The application addresses the need for a non-invasive and field-deployable screening tool capable of producing multi-class screening results supported by visual model explanations without relying on internet connectivity or immediate access to veterinary services. Rather than functioning as a substitute for

professional veterinary diagnosis, Scanine is intended to serve as a preliminary screening tool that presents classification results together with veterinarian-reviewed first-response recommendations that may be followed while veterinary consultation is being sought.

The study follows a descriptive and developmental research design that guides the design, implementation, and evaluation of the proposed application under conditions representative of community-based use. The system employs a MobileNetV3 model enhanced with Coordinate Attention and deployed through TensorFlow Lite within a Flutter-based mobile application to perform offline multi-class classification of canine skin images across four output categories: dermatophytosis, hotspot, flea allergy dermatitis, and dermatologically normal skin. To support the interpretation of classification results, the application generates Gradient-weighted Class Activation Mapping (Grad-CAM) heatmaps that highlight the image regions influencing the model's prediction. Following each classification, the application presents condition-specific first-response recommendations derived from veterinarian-reviewed guidance. Before classification, an on-device image quality assessment evaluates blur and lighting conditions and prompts users to recapture images that do not satisfy the application's predefined quality requirements. The application is intended for use by community responders conducting preliminary canine dermatological screening within Tacloban City.

1.3 Objective of the Study

This study focuses on developing Scanine, an offline mobile application that screens canine dermatological conditions through image classification, Grad-CAM interpretability, and veterinarian-reviewed first-response guidance, using smartphone images for community responders in Tacloban City.

Specifically, this study aims to:

1. Describe canine dermatological conditions through four-class image classification (dermatophytosis, hotspot, flea allergy dermatitis, dermatologically normal) using MobileNetV3 + Coordinate Attention, measured in terms of validation accuracy, precision, recall, and Area Under the Curve (AUC);
2. Determine the image regions influencing each classification using Grad-CAM, measured through correspondence with verified lesion locations;
3. Determine and validate condition-specific first-response recommendations for each condition, assessed for clinical accuracy by a licensed veterinarian;
4. Evaluate the system's functional integrity and software quality through:

    a. White-box testing using flutter_test and mocktail, to determine functional correctness, performance efficiency, and reliability of core functions; and

    b. Black-box testing, to assess Functional Suitability and Usability under ISO/IEC 25010, using a five-point Likert scale with veterinary professionals and community responders.

1.4 Scope and Limitation

This study focuses on the design, development, and preliminary evaluation of Scanine, a mobile-integrated screening application for multi-class canine dermatological screening, deployed as an offline mobile application built in Flutter for use by community responders conducting field screenings within Tacloban City. The system accepts standard smartphone images as its primary input and produces three outputs: a multi-class classification result identifying the detected condition as dermatophytosis, hotspot, flea allergy dermatitis, or dermatologically normal skin; a Grad-CAM activation heatmap that spatially localizes the image regions most influential to the model's decision; and condition-matched first-response recommendations retrieved from a locally stored, veterinarian-reviewed knowledge base. Field evaluation is confined to dog populations within Tacloban City, and the system's performance on populations from other geographic regions or with significantly different breed compositions has not been assessed within the scope of this study.

The application is restricted to the assessment of canine subjects and is not validated for use on felines or other animal species, given the species-specific dermatological variations that would require separate model training and clinical validation. The classification model is calibrated for high sensitivity to minimize the risk of missed detections, which may result in the flagging of visually similar conditions for professional veterinary verification, for instance, dermatophytosis lesions may overlap visually with early hotspot presentation, and flea allergy dermatitis may resemble other allergic or parasitic dermatoses. The interim care recommendations generated by the application are strictly informational and are

intended only as general first-response guidance for non-expert users; they do not constitute a definitive veterinary diagnosis, a treatment protocol, or a recommendation for specific pharmacological agents. All outputs are intended strictly as preliminary screening indicators. Accurate image processing is contingent on adequate ambient lighting and minimal motion blur, and full offline functionality is available once the application and its bundled classification model have been installed on the device, requiring no ongoing internet connection thereafter.

CHAPTER 2

This chapter reviews existing literature and systems to provide a comprehensive foundation, identifying current research gaps and technical benchmarks relevant to the study's execution.

Review of Related Literature and Systems

Review of Related Literature

2.1 Clinical Profile of Target Canine Dermatological Conditions

Global zoonotic health initiatives underscore the significant role of domestic and stray animals as primary reservoirs for various infectious diseases. Khaliq et al. (2026) emphasize that zoonotic parasites in stray dogs pose serious public health implications, particularly in areas where veterinary resources are scarce. Paryuni et al. (2020) further note that dermatophytosis, a prevalent fungal infection, remains a critical concern for companion animals worldwide due to its cross-species transmission potential. These global trends highlight a growing necessity for early screening mechanisms to protect both animal and human health in shared environments.

The clinical identification of these conditions is often complicated by visual similarities and overlapping symptoms. Mahendra (2025) explains that current perspectives on canine and feline dermatophytosis require a nuanced diagnostic approach because clinical presentations frequently mimic other dermatoses. Moriello et al. (2025) advocate for standardized clinical consensus guidelines to improve diagnostic accuracy and treatment outcomes in veterinary practice. These theoretical insights suggest that without objective screening tools, even experienced observers may struggle to differentiate between infectious and non-infectious skin lesions in the field.

Hypersensitivity and inflammatory conditions contribute another layer of complexity to the dermatological landscape of canine populations. Kearley (2025) identifies Flea Allergy Dermatitis (FAD) as a leading cause of chronic dermatological distress, requiring strict management protocols to alleviate animal suffering. Racine (2025) discusses the urgency of treating acute moist dermatitis, or "hotspots," noting that these lesions can expand rapidly without immediate intervention. Furthermore, Coutinho et al. (2023) provided detailed guidelines for allergen identification in atopic dermatitis, illustrating the variety of clinical factors that must be considered during assessment.

The prevalence of these conditions among stray dogs in the Philippines represents a localized public health challenge. Salces et al. (2022) conducted research across selected Philippine communities, finding significant risk factors for zoonotic parasites among roaming dog populations. This research confirms that urban centers face a persistent risk from untreated stray animals who serve as vectors for dermatological and parasitic infections. The lack of structured monitoring for these animals in communal spaces exacerbates the potential for zoonotic spread within local neighborhoods.

A survey by Inson et al. (2021) regarding canine skin cases in Metro Manila indicates that fungal and parasitic infections are the most frequent causes of veterinary consultation. Despite this high incidence, many community handlers and pet owners lack the objective training needed to distinguish these conditions during early stages. This documented clinical gap underscores a requirement for accessible screening methods that align with the most prevalent dermatoses found in local environments. Understanding these clinical profiles provides the essential ground truth for developing automated classification objectives.

2.2 Smartphone-Based Image Acquisition and Quality Assessment for Veterinary Screening

The transition of veterinary diagnostics toward mobile-based platforms relies heavily on the quality of input data collected by non-expert users. Taufiqoh and Purnamasari (2025) observe that image processing techniques are now at the center of developing detection models for skin diseases in pets. Biju et al. (2024) argue that while various machine learning approaches have matured, the integrity of the captured image remains the primary factor in classification reliability. This theoretical consensus suggests that an automated system must act as a quality filter before any deep learning inference occurs.

Capturing diagnostic-quality images in uncontrolled environments presents unique technical hurdles compared to laboratory settings. Hwang et al. (2022) explored multispectral imaging for dog skin disease but acknowledged that such specialized hardware is often too expensive for community use. Zou et al. (2026) similarly researched multi-spectral systems while emphasizing that standard smartphone optics remain the most practical platform for community screening. Consequently, the research community focuses on optimizing standard RGB images to achieve professional-grade results on consumer devices.

Recent literature highlights the importance of addressing environmental factors such as motion blur and poor lighting during image acquisition, with Newaz et al. (2025) arguing that accessible dermatology requires models that can handle mobile-acquired images with varying resolutions and focal qualities. Automated image quality checks are vital for identifying bacterial and fungal cases correctly in a mobile environment, preventing the processing of substandard captures that would otherwise lead to inaccurate classifications.

The theoretical justification for automated quality assessment is grounded in the need for user independence in field settings. Sri et al. (2025) highlight that pet skin problem detection is only as reliable as the user's ability to capture a clear, well-lit photograph. Kamath and Amrutha (2025) support this by proposing deep learning approaches that specifically account for the varied capture conditions found in mobile dermatology. The consensus in the field is that without an automated feedback loop, the reliability of a deep learning classifier is significantly diminished for non-experts.

A notable gap exists regarding the performance of these filters in the specific humid and outdoor environments found in tropical regions. Most studies are conducted in temperate or indoor settings where lighting and focus are more predictable for the sensor. There is a clear need for quality assessment logic designed specifically for the challenging field conditions encountered by community responders. This ensuring that responders can collect valid data regardless of their technical photography skills or environmental variables.

2.3 Lightweight Deep Learning Architectures and Attention Mechanisms for Mobile Veterinary AI

Deep learning has revolutionized medical imaging, but deploying these models on mobile hardware requires specialized architectural optimization rather than a direct transfer of research-grade designs. Patel and Vishwakarma (2025) discuss the development of automated detection for dog skin diseases using robust convolutional neural network backbones, demonstrating that CNN-based approaches can reliably distinguish between visually similar dermatological presentations. Building on this direction, Manoj et al. (2025) evaluated the ResNet50 architecture for this purpose, confirming its high classification accuracy in research settings where computational

resources and processing time are not primary constraints. However, these larger models often require significant computational resources, including substantial memory and processing power, that are unavailable on standard mobile devices used by the public. This gap between research-optimized accuracy and real-world deployment feasibility underscores the need for lightweight architectures capable of maintaining comparable diagnostic performance while operating within the hardware and battery limitations of consumer smartphones.

To address these resource constraints, the research community has pivoted toward lightweight architectures designed for efficient mobile execution. Ang et al. (2024) introduced "PetVision," which utilizes the MobileViTv3-Small architecture for animal breed classification, proving its efficiency. Upadhyay et al. (2023) successfully applied CNNs specifically for mobile dog skin disease identification, stressing the need for high-speed performance. Similarly, Madhavi et al. (2025) explored multi-label classification using ResNet18 to reduce the overall computational footprint without sacrificing accuracy.

The integration of attention mechanisms represents a recent innovation to improve the spatial sensitivity of these lightweight models. Jia et al. (2023) demonstrated through "MobileNet-CA-YOLO" that Coordinate Attention (CA) significantly improves feature extraction for small, localized anomalies. Attri et al. (2025) further proved that attention-based frameworks like "SAAM-VetNet" can accurately grade disease severity in animal populations. These theoretical advancements provide a basis for enhancing MobileNetV3 with CA to better identify localized dermatological lesions.

Comparative studies between standard mobile models and attention-augmented versions show clear performance improvements for medical tasks. Hardi et al. (2025)

noted that while transfer learning with VGG16 is effective, it consumes significantly more memory than optimized MobileNet-based alternatives. Alanazi (2025) argues that advanced CNN technology must be made field-deployable to have a true impact on veterinary care. The literature supports the choice of lightweight backbones due to their balanced hardware-aware design and efficient processing.

Despite these global advances, there is a lack of research focusing on the combination of MobileNetV3 and CA for specialized canine conditions. Most attention-based models are applied to human dermatology or agricultural pests rather than focused veterinary triage for stray populations. There is a specific opportunity to utilize the CA mechanism to refine the detection of ringworm, hotspots, and flea allergy dermatitis. This theoretical improvement aims to provide high-sensitivity screening within a native mobile architecture tailored for community use.

2.4 Explainable Artificial Intelligence through Gradient-Weighted Class Activation Mapping (Grad-CAM)

The "black-box" nature of deep learning has led to a global demand for Explainable AI (XAI) in medical applications. Jalaboi et al. (2022) proposed "DermX," a framework for explainable diagnosis, to help users understand the evidence behind a model's conclusion. Munjal et al. (2024) developed "SkinSage XAI," highlighting that transparency is critical for building user trust in diagnostic support tools. These global trends underscore that explainability is no longer optional for high-stakes veterinary screening.

Grad-CAM has emerged as a primary method for providing this transparency in visual AI systems. Campana et al. (2023) utilized Grad-CAM for skin lesion detection from smartphone images, proving that influential pixels can be highlighted

effectively for the user. Idroes et al. (2024) also applied this technique to skin lesion detection, reinforcing its utility in dermatological classification. By visualizing the "reasoning" of the network, users can verify that the AI is focusing on the actual lesion rather than background noise.

The application of XAI in digital health specifically helps non-expert users interpret classification results. Badhon et al. (2025) discuss using Grad-CAM alongside transfer learning to improve the accuracy of contour identification. Revathy et al. (2025b) specifically described a "MobileNet-Grad-Cam" framework for dermatological classification, aligning with the need for interpretability in lightweight models. This visualization is critical for community handlers who may be skeptical of an automated assessment without visual proof of the model's focus.

From a theoretical perspective, explainability improves the overall integrity and transparency of the diagnostic pipeline. Natarajan et al. (2026) argue that model explainability through Grad-CAM enhances the diagnostic process by allowing for human-in-the-loop verification. Shafiq et al. (2026) further discuss how explainable deep learning frameworks lead to more intelligent and robust classification systems. The literature suggests that a visual heatmap is an effective way to communicate AI logic to the public without overwhelming them.

A gap remains in delivering XAI to community responders in field settings. Nguyen et al. (2026) explored adaptive decision fusion, but few systems simplify XAI output for quick triage. Grad-CAM heatmaps should serve as a supporting visual aid, not a complex diagnostic map, keeping interpretability accessible in the field.

2.5 Native Mobile Edge AI and On-Device Inference Frameworks

The shift toward Edge AI represents a global movement to bring intelligence directly to the user's location for immediate action. Liu and Suzuki (2025) describe the theoretical benefits of Edge-AI for providing accurate and interpretable skin lesion diagnosis locally at the source. Nasir (2025) developed "DermAssist," a mobile system that alerts users in real-time, demonstrating the power of on-device intelligence. These studies establish that local inference is the most effective way to deploy diagnostic tools in areas with limited internet connectivity.

The choice of development frameworks determines a system's resilience and hardware efficiency in varied field settings. Martins et al. (2024) validate the use of Flutter and TensorFlow Lite for native mobile medical applications, citing their superior performance over web wrappers. Malawana et al. (2026) discussed systems relying on IoT connections but noted that persistent internet is a luxury in many underserved regions. Native development allows for deeper integration with mobile hardware compared to browser-based architectures.

Inference frameworks must be optimized to ensure that complex models do not cause latency on mobile devices. Kaviti (2025) emphasizes that while ensemble methods are powerful, they must be adapted for on-device convolutional networks to remain practical. Taranum et al. (2024) explored disease prediction patterns on mobile, showing that localized execution is key to user satisfaction. Chinta et al. (2024) successfully applied deep learning for specialized dog disease detection on mobile hardware using these optimized frameworks.

For field-based veterinary surveillance, an "offline-first" approach is the only viable strategy for reliable use in remote areas. Thoutam et al. (2023) discussed the

classification of dog skin disease in settings where internet infrastructure is intermittent or absent. Yashu et al. (2024) focused on enhancing canine dermatology through mobile image analysis in varied and challenging field conditions. These studies collectively support the move to a native inference framework to ensure consistent functionality during screenings.

Ultimately, the theoretical framework for mobile triage is built on the convergence of native development and edge AI. By using TensorFlow Lite, deep learning models can be optimized for high-speed, on-device execution on standard mobile hardware. This technical alignment fills a critical gap where most existing systems either require cloud access or suffer from high execution latency. A native, low-latency, and fully offline solution addresses the unique challenges of community-based field work.

Review of Related Systems

2.6 Multi-Class Canine Dermatological Screening Systems

Early veterinary systems often focused on professional clinical use rather than community-based triage for non-experts. Rathnayaka et al. (2022) developed an intelligent system for dog skin disease that used ontology-based extraction, but it required significant user expertise. Jose et al. (2025) introduced the "Canine Dermal Analyser," which successfully utilized AI for classification but was not designed for stray animal populations. These systems establish the potential for AI in dermatology but highlight a gap in accessibility for non-expert users.

In the Philippines, mobile pet management systems have focused more on documentation than automated diagnosis. The "Pet-Care" system by Baluran et al. (2023) provided a comprehensive management tool for pet owners but lacked an

integrated AI classification engine. While this local system improved pet wellness tracking, it still relied on the owner's visual judgment for identifying health concerns. There is a need for systems that integrate a deep learning engine to provide objective screening for common local conditions.

The progression of these systems shows a clear requirement for tools that can handle multiple conditions simultaneously in the field. Existing systems often struggle to balance comprehensive diagnostic output with a simplified mobile interface for volunteers. A targeted multi-class approach ensures that a system is both specialized and accessible to community handlers who encounter varied dermatoses. Such an approach bridges the gap between general pet management and professional veterinary diagnosis.

2.7 Mobile Deep Learning Classification Systems with Attention Mechanisms

The integration of attention modules has set a new standard for mobile classification accuracy in specialized fields. Jia et al. (2023) demonstrated through "MobileNet-CA-YOLO" that attention mechanisms significantly improve the detection of localized defects in agriculture. Ang et al. (2024) also used a MobileViTv3-based architecture for the "PetVision" system, proving its efficiency in animal classification tasks. These systems show that lightweight models can achieve high integrity when augmented with spatial sensitivity layers.

However, most veterinary classification systems still rely on standard CNN backbones without attention modules. Harshitha et al. (2024) and Patel and Vishwakarma (2025) both used standard CNN or broad deep learning approaches without incorporating attention mechanisms to refine feature extraction. Literature

suggests that combining MobileNetV3 with Coordinate Attention is necessary to capture the subtle visual markers of canine lesions.

This architectural improvement represents a shift from general classification to high-precision dermatological screening on mobile devices. Previous systems have established the feasibility of mobile deep learning but often suffer from accuracy trade-offs. By including a Coordinate Attention block, a system can address the spatial sensitivity issues identified in standard MobileNet deployments. This justifies the technical revision as a necessary step for professional-grade screening in a lightweight mobile package.

2.8 Explainable AI-Based Veterinary Diagnostic Support Systems

Explainability is a common feature in recent human medical systems, yet it remains underutilized in the community veterinary sector. Systems such as "SkinSage XAI" (Munjal et al., 2024) and "DermX" (Jalaboi et al., 2022) have set the standard for transparent AI in dermatology. These systems use visualization to explain the model's logic, which is essential for building confidence in the AI's results. Despite their success, these systems are rarely adapted for field use by non-expert animal handlers in rural areas.

Existing Grad-CAM implementations, such as those by Revathy et al. (2025b) and Natarajan et al. (2026), show the potential for visual interpretability on mobile devices. However, many of these systems do not link the visual heatmap to specific first-response guidance for the user. An improved system would pair Grad-CAM visualizations with veterinarian-reviewed care recommendations in a single interface. This ensures that the user understands the model's findings and knows exactly what the next steps should be.

These related systems demonstrate that XAI is most effective when it is part of a larger decision-support framework. Simply identifying a lesion is insufficient if the user does not trust the result or know how to act on it safely. By focusing on explainability, a system bridges the gap between machine intelligence and human intervention. This approach is particularly valuable for volunteers who are dealing with stray animals and need immediate confirmation to proceed with care.

2.9 Native Mobile Edge AI Screening Systems

The debate between web-based and native deployment has shaped the development of mobile diagnostic tools over recent years. The system by Malawana et al. (2026) relied on IoT and persistent internet connections, which limits its usefulness in remote areas. In contrast, the native "X-Ray Covid19" application (Martins et al., 2024) demonstrated the robustness of Flutter and TFLite for offline medical triage. This system serves as a direct technical predecessor for native mobile architectures in screening.

A critical argument in mobile development is that native edge AI is superior to web-based solutions for veterinary field triage. While web apps are easier to deploy, native apps offer better performance for deep learning and more reliable offline access to hardware. By choosing a native TFLite-powered framework, a system improves inference speed and overall system stability. This addresses the technical gap identified in earlier web-based veterinary monitoring systems that struggled with performance on mid-range hardware.

Furthermore, the native approach allows for the implementation of advanced image quality filters that run before the model is invoked. Liu and Suzuki (2025) describe how edge-aware features are better preserved when processed natively at the

device level. This architectural choice ensures that an application remains functional in outskirts where cloud connectivity is not guaranteed. A native deployment represents a commitment to reliability and performance in the most challenging field conditions.

2.10 AI-Assisted First-Response and Community Veterinary Support Systems

Community animal health tools have traditionally focused on documentation rather than immediate decision support for responders. Systems like "DermAssist" (Nasir, 2025) provide automated alerting but are not tailored for the specific zoonotic risks of stray dog populations. The clinical consensus guidelines discussed by Moriello et al. (2025) emphasize the need for safe, expert-reviewed guidance, which is often missing from purely automated tools. A system addresses this by providing static, expert-reviewed first-response protocols for each classification.

Other systems, such as the demodicosis detection tool by Chinta et al. (2024), focus purely on the model's accuracy rather than the user's next steps. While high accuracy is important, it does not solve the problem of providing immediate care in the field. Rathnayaka et al. (2022) attempted to provide clinical extraction, but the results were too complex for non-veterinary users to act upon safely. A simplified process delivers actionable, static care guidance that prioritizes the welfare of the stray animal.

This community-centric approach is validated by local research into the zoonotic risks of stray dog populations. Khaliq et al. (2026) and Salces et al. (2022) have both emphasized the need for community-level intervention to manage animal-borne diseases. By providing AI-assisted first-response guidance, a system empowers animal welfare volunteers to act as a first line of defense. The final argument for such a system is its ability to translate complex AI classification into safe, actionable community veterinary support.

CHAPTER 3

This chapter presents the technical framework of Scanine, outlining the system's architecture, workflow, and core components. It examines the methods and technologies employed, identifies existing limitations in current canine dermatological screening practices, and explains the design strategies implemented to address these constraints while maintaining reliability and feasibility.

Technical Background

Scanine is built on lightweight deep learning, on-device native inference, explainable AI, and offline-capable mobile architecture, each selected for its fitness to the constraints of field deployment on consumer-grade Android smartphones without internet connectivity. The sections that follow detail the current screening practices the system is designed to supplement, the proposed system and its workflow, and the software, hardware, and peopleware requirements underpinning its development.

3.1 Current System

The current approach to canine dermatological screening in community and field settings is largely observational and dependent on the availability of veterinary personnel. In practice, dermatophytosis is identified through visual inspection of the animal's coat and skin, with confirmation typically requiring a Wood's lamp examination, fungal culture, or microscopic examination of hair shafts conducted under laboratory conditions. Hotspots, or acute moist dermatitis, are diagnosed through direct clinical assessment of the lesion's presentation, with cytology used in ambiguous cases to rule out secondary infections or overlapping dermatoses. Flea allergy dermatitis, meanwhile, is confirmed through a combination of visual inspection, flea combing, intradermal allergen testing, and therapeutic elimination trials, none of which

are feasible outside a clinical setting. Across all three conditions, the diagnostic process demands physical proximity to the animal, specialized equipment, and trained personnel, resources that are consistently unavailable to the non-specialist community responders most likely to encounter affected animals first. Animals displaying visible lesions may go unexamined for extended periods, and without a standardized screening mechanism, decisions about whether to escalate a case for veterinary referral depend entirely on the subjective judgment of whoever encounters the animal.

Beyond the immediate diagnostic limitations, the broader surveillance infrastructure for these conditions in community settings is largely reactive. Cases are typically escalated only after visible signs have progressed to a stage unambiguous to an untrained observer, a delay that carries distinct public health consequences for each condition. Dermatophytosis poses a direct zoonotic transmission risk to humans, particularly children and immunocompromised individuals in communities with close animal contact. Hotspots, if left unmanaged, can expand rapidly into extensive bacterial wounds through continued self-trauma. Flea allergy dermatitis perpetuates flea burden across the local animal population, compounding re-exposure risk for both animals and people sharing the same environment. There is currently no community-level mechanism for early, standardized detection across these presentations, no tool capable of distinguishing between them without specialist input, and no field-deployable system that can convert an image of a suspect lesion into an actionable, condition-specific output without laboratory support. The result is a surveillance gap at the human-animal interface that existing clinical methods were not designed to fill.

3.2 Proposed System

Scanine operates as an offline mobile application built in Flutter, designed to assist community responders in screening dogs for three canine skin conditions, dermatophytosis, acute moist dermatitis (hotspot), and flea allergy dermatitis, through standard smartphone image capture alone. Users point the device's rear-facing camera at the animal, and the system captures five frames per screening event. Each frame is checked for image quality, then classified individually, and the five results are combined through majority vote into one final classification. If that combined result isn't confident enough, the system won't assign a class. Instead, it returns an inconclusive result and tells the user to consult a veterinarian. When a class is confidently assigned, the application generates a Gradient-weighted Class Activation Mapping (Grad-CAM) heatmap that spatially localizes the regions of pathological skin texture identified by the model for the detected condition. Upon classification, the application generates a Gradient-weighted Class Activation Mapping (Grad-CAM) heatmap that spatially localizes the regions of pathological skin texture identified by the model for the detected condition. Simultaneously, condition-matched first-response recommendations are retrieved from a knowledge base bundled locally with the application and delivered alongside the heatmap, providing interim guidance calibrated to the specific condition detected. These three outputs, the multi-class classification result, the Grad-CAM heatmap, and the condition-specific first-response recommendations, are presented together as a single screening report intended to support, rather than replace, the decision to pursue veterinary referral and to guide interim care in the meantime. The system does not require an internet connection during operation once installed, and it is designed for use by non-specialist community responders without prior technical training.

3.2.1 Software

The software stack provides the infrastructure necessary for on-device inference, real-time image processing, explainability generation, condition-specific guidance delivery, and offline application availability. Table 3.1 outlines the primary software requirements and their functional significance in the system architecture.

Table 3.1. *System Software Requirements*

Software Requirement	Specification	Rationale
Flutter	Cross-platform UI toolkit (Dart), widget-based architecture	Builds the mobile application interface, managing camera integration, result display, and application state across user interactions
TensorFlow Lite (tflite_flutter)	On-device machine learning inference plugin	Executes the quantized MobileNetV3 model enhanced with Coordinate Attention directly on the device, enabling offline inference without server-side processing
image (Dart package)	Pure-Dart image processing library	Performs on-device blur detection through Laplacian variance analysis and lighting/exposure checks on captured frames prior to inference
Git / GitHub	Version control and collaborative code management	Manages codebase versioning and tracks development changes across the project timeline
Visual Studio Code	Integrated Development Environment	Provides the primary coding environment for writing, testing, and debugging the system's modular components

The software architecture is designed around the constraint of full offline operability on a consumer-grade Android smartphone. Flutter serves as the application framework, managing the interface, camera integration, and screen navigation across the screening workflow. TensorFlow Lite, accessed through the tflite_flutter plugin, handles all model inference on-device, eliminating any dependency on a server or persistent internet connection during operation. Before a captured frame is submitted to the classifier, the image package performs Laplacian variance analysis to detect motion blur and flags frames with inadequate lighting, prompting the user to recapture the image if it does not meet the application's quality threshold. Frames that pass this filtering step are submitted to the quantized MobileNetV3 model enhanced with Coordinate Attention for multi-class evaluation. Upon a classification result, the Grad-CAM module computes activation maps from the model's final convolutional layer, rendered as heatmap overlays that spatially localize the pathological region associated with the detected condition. Condition-matched first-response recommendations, reviewed by a veterinary professional, are then retrieved from a knowledge base bundled locally with the application and delivered as part of the unified screening output. Because the classification model, the image quality filter, and the first-response knowledge base are all packaged within the application at install time, the complete screening workflow executes under full network disconnection once installation is complete. Git and Visual Studio Code support the development environment without contributing to the runtime architecture of the deployed application.

3.2.2 Hardware

The hardware requirements reflect the system's design priority of operability on widely available consumer devices rather than specialized equipment. Table 3.2 specifies the hardware components considered in the development and deployment of the system.

Table 3.2. *System Hardware Requirements*

Hardware Requirement	Specification	Rationale
Laptop / PC	Intel Core i5 or equivalent, minimum 8GB RAM, 256GB SSD storage, Windows 10 or later	Supports local development, model training, post-training quantization, and full application testing within the development environment
Target Device	Android smartphone, Android 8.0 (Oreo) or later, minimum 3GB RAM, rear-facing camera with autofocus	Serves as the primary field deployment hardware; the system executes inference and renders outputs natively on-device following installation, without requiring a browser or additional runtime dependencies

The hardware selection reflects the deployment reality of community-level field screening, where access to specialized computing equipment cannot be assumed. Development is conducted on a standard laptop or desktop machine with sufficient processing capacity to support model training, quantization, and local application testing. The target deployment hardware is a mid-range Android smartphone, the device type most accessible to community responders operating in Tacloban City, with the rear-facing camera serving as the sole image acquisition input. The system's use of

a quantized MobileNetV3 model with Coordinate Attention, executed through TensorFlow Lite, is directly motivated by this hardware constraint, ensuring that inference latency and memory consumption remain within the operational limits of mid-range devices without dedicated neural processing units or high-capacity RAM.

3.2.3 Peopleware

Scanine is designed to accommodate two distinct user groups, each with a defined role in the screening workflow. System Administrators are responsible for the foundational management of the application, including maintaining the bundled classification model and first-response knowledge base content, preparing and distributing application updates, and any configuration changes required to adapt the system to new deployment contexts. The primary end-users are community responders, animal welfare officers, barangay health workers, and other non-specialist personnel engaged in canine health monitoring within Tacloban City. These users perform the core operational function: capturing images of suspected dermatological cases in the field, reviewing the multi-class classification result, Grad-CAM heatmap, and condition-specific first-response recommendations, and using these outputs to decide whether to escalate a case for professional veterinary assessment and what interim measures to apply before that referral occurs. No prior technical training or clinical expertise is required, as the system produces interpretable outputs designed to be actionable without specialist knowledge.

CHAPTER 4

This chapter presents the methodology and systematic development of Scanine. It describes the functional and non-functional requirements, the system architecture, data handling procedures, and implementation processes, providing a structured account of the system's progression from conceptual design to a functional multi-class canine dermatological screening tool.

Methodology, Results, and Discussion

4.1 Requirements Analysis

Requirements analysis identifies what Scanine must achieve to function correctly and satisfy user needs. This section examines the key expectations for the system, focusing on three main areas: performance, safety, and security, which collectively ensure that the system delivers accurate, reliable, and actionable screening outputs.

4.1.1 Performance Requirements

Performance requirements define how the system should handle image processing workloads, execute inference efficiently, and respond in a timely manner under field conditions. They ensure that the application can perform real-time multi-class classification and produce outputs without errors or unacceptable delays on mid-range Android hardware.

Table 4.1. *Performance Requirements*

Performance Requirement	Description	Rationale
On-Device Inference Speed	The system shall complete model inference and produce a classification	Ensures that the screening workflow remains responsive in

	output within an acceptable latency window per image on mid-range Android devices.	the field without requiring the responder to wait between captures.
Bundled Asset Availability	All application assets, including the quantized classification model and the first-response knowledge base, shall be packaged with the application at install time and available immediately after installation.	Ensures that the system operates without degradation in environments with no network access, since no post-install download is required.
Image Quality Filter Responsiveness	The blur and brightness detection filter shall evaluate each captured image in real time and return a pass or reject decision before the image reaches the inference pipeline.	Prevents low-quality images from entering the classification process, preserving the integrity of outputs without introducing perceptible delay.

The performance requirements establish the technical benchmarks for reliable and responsive operation under field conditions. Central to these requirements is On-Device Inference Speed, which ensures that the quantized MobileNetV3 model, enhanced with Coordinate Attention and executed through TensorFlow Lite, produces multi-class classification outputs within an acceptable response window, maintaining a usable experience on hardware without dedicated neural processing units. Bundled Asset Availability ensures that the classification model and first-response knowledge base are packaged within the application at install time, so that full functionality is available immediately after installation without any dependency on a later network connection. Image Quality Filter Responsiveness ensures that the on-device blur and

brightness detection, implemented through the image Dart package, evaluates each captured image in real time, rejecting unsuitable inputs before they reach the model so that data integrity is maintained without introducing noticeable delays for the field responder.

4.1.2 Safety Requirements

Safety requirements focus on maintaining output accuracy and system reliability. They are implemented to prevent unsuitable inputs from reaching the classification pipeline, ensure that outputs are interpretable and appropriately qualified, and protect the integrity of the system's screening results, safeguarding responders from making triage decisions based on unreliable information.

Table 4.2. *Safety Requirements*

Safety Requirement	Description	Rationale
Image Quality Gate	The system shall reject any camera frame that fails the Laplacian variance blur detection or brightness threshold before it is passed to the classification model.	Prevents low-quality inputs from generating unreliable classification results, preserving the validity of downstream outputs including the Grad-CAM heatmap and condition-specific first-response guidance.
Confidence Abstention Gate	The system shall withhold a class assignment when it isn't confident in the result, returning an inconclusive outcome that tells the user to consult a veterinarian instead.	Stops the system from confidently guessing on conditions it wasn't trained to recognize, such as mange or pyoderma, turning a hidden misclassification into a clear referral.

Output Qualification	The system shall explicitly label all screening outputs as preliminary triage indicators and shall not present results as definitive veterinary diagnoses or pharmacological recommendations.	Acknowledges the inherent limitations of automated image-based screening, ensuring that responders treat outputs as prioritization guidance rather than clinical conclusions.
Species Restriction Notice	The system shall display a notification clarifying that the application is calibrated for canine subjects only and has not been validated for use on other animal species.	Prevents misapplication of the model on feline or other animal subjects whose dermatological presentations differ from those in the training dataset.
Guidance Output Qualification	The system shall present condition-specific first-response recommendations as informational interim care guidance only, and shall not present these recommendations as definitive treatment protocols or pharmacological prescriptions.	Ensures that the guidance output is interpreted as a non-clinical, community-level triage aid intended exclusively for use prior to professional veterinary consultation.

By implementing the Image Quality Gate, the system addresses the primary challenge of field image collection: the high variability in lighting and motion blur that characterizes uncontrolled outdoor environments. This ensures that classification outputs are built on input data that meets a minimum quality threshold rather than on degraded frames that would otherwise produce unreliable results.

The qualification requirements collectively define the responsible boundary of the system's outputs. By explicitly labeling classification results as preliminary triage indicators, qualifying condition-specific first-response guidance as informational and non-clinical, and restricting application scope to canine subjects, the system ensures

that responders at all technical levels understand what the outputs are and are not, providing a clear boundary that supports the validity of any triage decision that follows.

4.1.3 Security Requirements

Security requirements define how the system maintains the integrity of the screening workflow and protects the application from misuse. They include measures to ensure correct asset delivery, prevent unauthorized modification of bundled components, and handle user data responsibly within the constraints of an offline-first mobile application.

Table 4.3. *Security Requirements*

Security Requirements	Description	Rationale
Bundled Asset Integrity	The system shall verify the integrity of the bundled classification model and first-response knowledge base files at application startup through checksum validation.	Prevents a corrupted or tampered model file from being loaded into the inference pipeline, ensuring that classification outputs are produced by the validated model version.
Compliance with Data Privacy Laws	The system shall adhere to applicable data privacy regulations, including the Data Privacy Act of 2012 (RA 10173) of the Philippines, ensuring that all user interactions and screening records are handled in accordance with legal and ethical standards.	Promotes lawful and ethical handling of screening data, protects the rights of individuals whose animals are screened, and reduces legal liability risk.

Screening Data Confidentiality	All locally stored screening records shall be stored within an encrypted local Hive database accessible only within the application context, and shall not be transmitted to external servers without explicit user action.	Preserves the confidentiality of field screening data in communities where sensitive animal health information may have implications for public health reporting or individual privacy.health reporting or individual privacy.

The security architecture of Scanine is designed around the constraint that the system operates primarily offline, without server-side authentication or centralized data management. Bundled Asset Integrity ensures that both the classification model and the first-response knowledge base are validated against known checksums at application startup, preventing a corrupted or outdated asset from being silently used in place of the validated build. Compliance with RA 10173 and the principle of screening data confidentiality together ensure that locally stored records remain within an encrypted Hive database local to the device, providing a responsible baseline for data handling even in the absence of a server-side security layer.

4.2 Design of Software, Systems, Product, and/or Processes

This section provides a detailed overview of the system's structural design and logic. It expounds on the specific Systems Development Life Cycle model adopted for the project and presents various diagrams to illustrate the data flow, functional processes, and their interactions.

Systems Development Life Cycle

The proposed system adopted the Sashimi Model in its development process. The Sashimi Model is a specialized evolution of the traditional Waterfall methodology, defined by its overlapping phase structure that allows for a more fluid and continuous transition between the different stages of software development. Unlike rigid linear models where one phase must conclude before the next begins, the Sashimi approach permits the development team to start a new phase while the previous one is still in progress, ensuring that insights discovered during implementation or design can be immediately fed back into requirements or architectural logic for a more integrated and responsive development environment (Kramer & Kramer, 2023).

For the proposed system, the Sashimi Model provides a bridge between the intersecting demands of machine learning development and mobile application engineering. Since the system must integrate a trained and quantized multi-class neural network with a Grad-CAM module and a first-response guidance retrieval pipeline into an on-device native inference environment, the overlapping phases allow the development team to refine the MobileNetV3 model with Coordinate Attention simultaneously while designing the Flutter-based mobile application interface. This ensures that Grad-CAM overlays and condition-indexed knowledge base entries are calibrated against actual model activation outputs and classification categories as the mobile screening workflow is being constructed, rather than integrated as an afterthought. By using this model, the project maintains technical coherence across its machine learning, explainability, guidance retrieval, and deployment components, ensuring that the final application functions as a unified screening tool.

The development cycle of the proposed system, following the Sashimi Model, is structured into the following phases: Planning, Designing, Development, Testing, and Implementation, as shown in Figure 4.1.

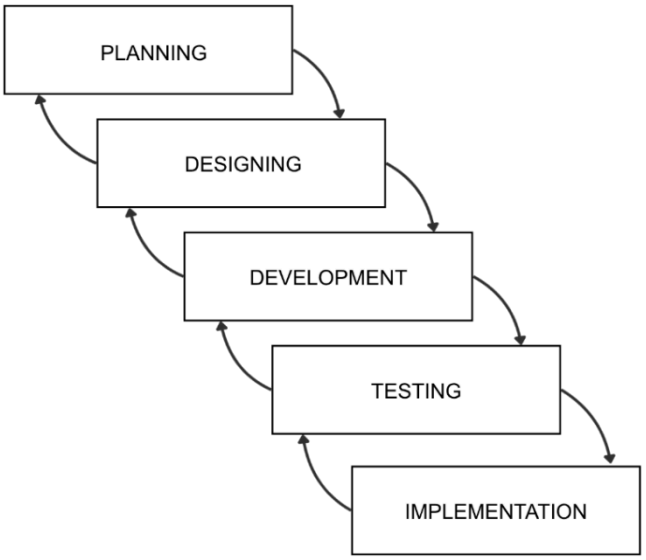


Figure 4.1. *Systems Development Life Cycle (Sashimi Model)*

Planning Phase

In this phase, preparatory activities were undertaken to establish a clear direction for the development of Scanine. This phase focused on identifying the essential requirements and organizing the sequence of development tasks to ensure a structured and manageable workflow, better supporting the system's objective of translating standard smartphone images of canine subjects into interpretable, condition-specific triage outputs. By clarifying what the system must accomplish and defining the foundational considerations for its operation, this phase provided a structured basis for the succeeding design, development, testing, and implementation stages.

Requirements Analysis

The system's requirements, as detailed in Section 4.1, cover performance, safety, and security considerations. Performance requirements ensure the system can execute real-time image quality filtering and complete on-device multi-class inference within an acceptable latency window on mid-range Android hardware. Safety requirements emphasize input quality control, output qualification, species restriction, and the appropriate framing of condition-specific first-response guidance as informational interim care rather than clinical treatment, safeguarding the validity of screening results and supporting responsible triage prioritization. Security requirements maintain system integrity through checksum validation of bundled model and knowledge base assets, and screening data confidentiality via an encrypted local Hive database in accordance with the Data Privacy Act of 2012, preserving the consistency and trustworthiness of the application across offline deployment instances.

Activity List

The activity list is a structured compilation of tasks necessary to develop the proposed system, detailing what needs to be done and in what sequence. It outlines the essential steps specific to the system, from project initiation and initial documentation to model training, application development, testing, and evaluation, providing a clear roadmap to ensure systematic, organized, and efficient development of the platform as shown in Table 4.4.

Table 4.4. *Activity List*

ID	ACTIVITY NAME	PREDECES-SOR(S)	DURATION (DAYS)	START DATE	END DATE
Phase1: Planning					
A	Proposal Conceptualization and Initial Planning	—	1	Jan 19, 2026	Jan 19, 2026
B	Official Adviser Assignment	A	3	Jan 20, 2026	Jan 22, 2026
C	Proposal Review and Approval	B	1	Jan 21, 2026	Jan 21, 2026
D	Literature Review Mapping and Gap Analysis	B	7	Jan 23, 2026	Jan 29, 2026
E	Stakeholder Coordination and Domain Consultation	B	5	Jan 23, 2026	Jan 27, 2026
F	Chapter 1 Drafting (Intro & Problem Context)	C, D	7	Jan 30, 2026	Feb 5, 2026
G	Chapter 1 Consultation and Revision	F	3	Feb 6, 2026	Feb 8, 2026
H	Rechecking of Chapter 1 Revisions	G	2	Feb 9, 2026	Feb 10, 2026
I	Chapter 1 Final Submission	H	1	Feb 11, 2026	Feb 11, 2026
J	Chapter 2 Drafting (Review of Literature)	D	7	Jan 30, 2026	Feb 5, 2026
K	Chapter 2 Consultation and Revision	J	3	Feb 6, 2026	Feb 8, 2026
L	Rechecking of Chapter 2 Revisions	K	2	Feb 9, 2026	Feb 10, 2026
M	Chapter 2 Final Submission	L	1	Feb 11, 2026	Feb 11, 2026
Phase 2: Designing					
N	Chapter 3 Drafting (Tech Background)	C	6	Jan 22, 2026	Jan 27, 2026

O	Veterinary and Domain Expert Consultation	E	5	Jan 28, 2026	Feb 1, 2026
P	Chapter 3 Consultation and Revision	N, O	3	Feb 2, 2026	Feb 4, 2026
Q	Rechecking of Chapter 3 Revisions	P	2	Feb 5, 2026	Feb 6, 2026
R	Chapter 3 Final Submission	Q	1	Feb 7, 2026	Feb 7, 2026
S	Algorithmic Pipeline Design	O	7	Feb 2, 2026	Feb 8, 2026
T	Multi-Class Dataset Collection Protocol and Annotation Specification	O	5	Feb 2, 2026	Feb 6, 2026
Phase 3: Development					
U	Canine Skin Conditions Image Collection	T	14	Feb 7, 2026	Feb 20, 2026
V	Dataset Cleaning and Deduplication	U	5	Feb 21, 2026	Feb 25, 2026
W	Multi-Class Image Labeling and Annotation	V	7	Feb 26, 2026	Mar 4, 2026
X	Data Augmentation Strategy and Implementation	W	5	Mar 5, 2026	Mar 9, 2026
Y	5-Frame Sequential Capture Module Development	S	6	Feb 9, 2026	Feb 14, 2026
Z	Blur Detection Module Development (Laplacian Variance)	Y	4	Feb 15, 2026	Feb 18, 2026
AA	Brightness Check Module Development (Mean Pixel Intensity)	Z	4	Feb 19, 2026	Feb 22, 2026
AB	Retake-Prompt UX Flow Development (Quality-Gate Feedback Loop)	Z, AA	4	Feb 23, 2026	Feb 26, 2026

AC	Tensor Conversion and Normalization Module Development	S	4	Feb 9, 2026	Feb 12, 2026
AD	Bilinear Resize and Batch Dimension Module Development	AC	3	Feb 13, 2026	Feb 15, 2026
AE	Preprocessing and Enhancement Integration Testing	AA, AD	4	Feb 23, 2026	Feb 26, 2026
AF	Coordinate Attention Module Design and Integration into MobileNetV3-Small Backbone	AE, X	6	Mar 10, 2026	Mar 15, 2026
AG	MobileNetV3-Small Multi-Class Training	AF	10	Mar 16, 2026	Mar 25, 2026
AH	Multi-Frame Voting Logic Development and Validation	AG	5	Mar 26, 2026	Mar 30, 2026
AI	Softmax Classification Head Integration and Validation	AH	4	Mar 31, 2026	Apr 3, 2026
AJ	Client App Development	S, AB	15	Feb 27, 2026	Mar 13, 2026
AK	AI Model Development — 50% Progress Milestone	AI, AJ	1	Apr 4, 2026	Apr 4, 2026
AL	Chapter 4 Drafting	Q	6	Feb 7, 2026	Feb 12, 2026
AM	Chapter 4 Consultation and Revision	AL	3	Feb 13, 2026	Feb 15, 2026
AN	Rechecking of Chapter 4 Revisions	AM	2	Feb 16, 2026	Feb 17, 2026
AO	Chapter 4 Final Submission	AN	1	Feb 18, 2026	Feb 18, 2026
Phase 4: Testing and Iterative Development					
AP	Pre-Oral Defense Presentation	AK	1	May 29, 2026	May 29, 2026
AQ	Post-Defense	AP	63	May 30,	Jul 31, 2026

	Manuscript Revisions			2026	
AR	Research Ethics Compliance Verification	AQ	1	Aug 5, 2026	Aug 5, 2026
AS	Post-Defense System Refinement	AP, AK	63	May 30, 2026	Jul 31, 2026
AT	First-Response Content Drafting and Veterinarian Review (Clinical Safety Sign-off)	O	10	Feb 2, 2026	Feb 11, 2026
AU	Grad-CAM Heatmap Generation Setup	AR, AG	10	Aug 6, 2026	Aug 15, 2026
AV	Output Synthesis, Decision Logic, and First-Response Guidance Integration	AU, AS, AR, AT	16	Aug 16, 2026	Aug 31, 2026
AW	End-to-End Core Integration Testing and On-Device Inference Speed Benchmarking	AV	17	Sep 1, 2026	Sep 17, 2026
AX	Standard System Testing, QA, and Offline Functionality Validation	AW	15	Sep 18, 2026	Oct 2, 2026
AY	Field Evaluation with Stray Dog Populations, Tacloban City — Accuracy, Speed, Usability	AX	10	Oct 3, 2026	Oct 12, 2026
Phase 5: Implementation					
AZ	Chapter 5 Drafting	AX	6	Oct 3, 2026	Oct 8, 2026
BA	Chapter 5 Consultation and Revision	AZ, AS	3	Oct 9, 2026	Oct 11, 2026
BB	Rechecking of Chapter 5 Revisions	BA	2	Oct 12, 2026	Oct 13, 2026
BC	Chapter 5 Final Submission	BB	1	Oct 14, 2026	Oct 14, 2026

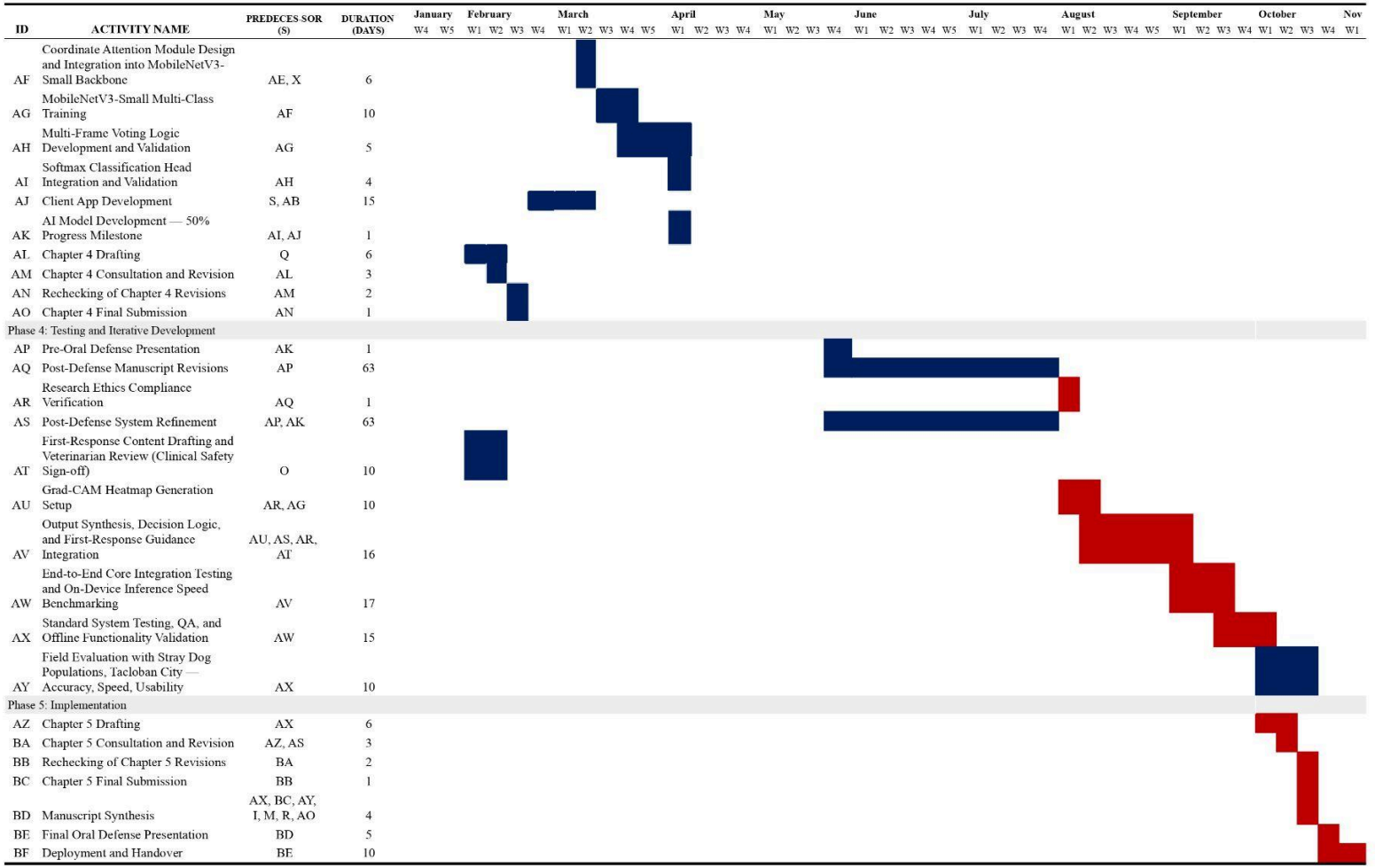
BD	Manuscript Synthesis	AX, BC, AY, I, M, R, AO	4	Oct 15, 2026	Oct 18, 2026
BE	Final Oral Defense Presentation	BD	5	Oct 19, 2026	Oct 23, 2026
BF	Deployment and/or Handover	BE	10	Oct 24, 2026	Nov 2, 2026

Gantt Chart

The Gantt chart provides a comprehensive visual timeline for Scanine, as shown in Table 4.5. It maps the concurrent workflows between dataset acquisition, model training, and application development to ensure an optimized progression toward the final oral defense in October 2026. By organizing the research into five overlapping phases via the Sashimi SDLC model, the chart translates Early Start and Late Finish parameters into a horizontal timeline, clearly distinguishing the critical path milestones from tasks that possess operational float.

Table 4.5. *GANTT Chart*

ID	ACTIVITY NAME	PREDECESSOR (S)	DURATION (DAYS)	January		February				March					April				May				June					July				August					September				October				Nov
				W4	W5	W1	W2	W3	W4	W1	W2	W3	W4	W5	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W5	W1	W2	W3	W4	W1	W2	W3	W4	W5	W1	W2	W3	W4	W1	W2	W3	W4	W1
Phase1: Planning																																												
A	Proposal Conceptualization and Initial Planning	—	1																																									
B	Official Adviser Assignment	A	3																																									
C	Proposal Review and Approval	B	1																																									
D	Literature Review Mapping and Gap Analysis	B	7																																									
E	Stakeholder Coordination and Domain Consultation	B	5																																									
F	Chapter 1 Drafting (Intro & Problem Context)	C, D	7																																									
G	Chapter 1 Consultation and Revision	F	3																																									
H	Rechecking of Chapter 1 Revisions	G	2																																									
I	Chapter 1 Final Submission	H	1																																									
J	Chapter 2 Drafting (Review of Literature)	D	7																																									
K	Chapter 2 Consultation and Revision	J	3																																									
L	Rechecking of Chapter 2 Revisions	K	2																																									
M	Chapter 2 Final Submission	L	1																																									
Phase 2: Designing																																												
N	Chapter 3 Drafting (Tech Background)	C	6																																									
O	Veterinary and Domain Expert Consultation	E	5																																									
P	Chapter 3 Consultation and Revision	N, O	3																																									
Q	Rechecking of Chapter 3 Revisions	P	2																																									
R	Chapter 3 Final Submission	Q	1																																								
S	Algorithmic Pipeline Design	O	7																																									
T	Multi-Class Dataset Collection Protocol and Annotation Specification	O	5																																									
Phase 3: Development																																												
U	Canine Skin Conditions Image Collection	T	14																																										
V	Dataset Cleaning and Deduplication	U	5																																										
W	Multi-Class Image Labeling and Annotation	V	7																																										
X	Data Augmentation Strategy and Implementation	W	5																																										
Y	5-Frame Sequential Capture Module Development	S	6																																										
Z	Blur Detection Module Development (Laplacian Variance)	Y	4																																										
AA	Brightness Check Module Development (Mean Pixel Intensity)	Z	4																																										
AB	Retake-Prompt UX Flow Development (Quality-Gate Feedback Loop)	Z, AA	4																																										
AC	Tensor Conversion and Normalization Module Development	S	4																																										
AD	Bilinear Resize and Batch Dimension Module Development	AC	3																																										
AE	Preprocessing and Enhancement Integration Testing	AA, AD	4																																										



ID	ACTIVITY NAME	PREDECESSOR (S)	DURATION (DAYS)
AF	Coordinate Attention Module Design and Integration into MobileNetV3-Small Backbone	AE, X	6
AG	MobileNetV3-Small Multi-Class Training	AF	10
AH	Multi-Frame Voting Logic Development and Validation	AG	5
AI	Softmax Classification Head Integration and Validation	AH	4
AJ	Client App Development	S, AB	15
AK	AI Model Development — 50% Progress Milestone	AI, AJ	1
AL	Chapter 4 Drafting	Q	6
AM	Chapter 4 Consultation and Revision	AL	3
AN	Rechecking of Chapter 4 Revisions	AM	2
AO	Chapter 4 Final Submission	AN	1
Phase 4: Testing and Iterative Development			
AP	Pre-Oral Defense Presentation	AK	1
AQ	Post-Defense Manuscript Revisions	AP	63
AR	Research Ethics Compliance Verification	AQ	1
AS	Post-Defense System Refinement	AP, AK	63
AT	First-Response Content Drafting and Veterinarian Review (Clinical Safety Sign-off)	O	10
AU	Grad-CAM Heatmap Generation Setup	AR, AG	10
AV	Output Synthesis, Decision Logic, and First-Response Guidance Integration	AU, AS, AR, AT	16
AW	End-to-End Core Integration Testing and On-Device Inference Speed Benchmarking	AV	17
AX	Standard System Testing, QA, and Offline Functionality Validation	AW	15
AY	Field Evaluation with Stray Dog Populations, Tacloban City — Accuracy, Speed, Usability	AX	10
Phase 5: Implementation			
AZ	Chapter 5 Drafting	AX	6
BA	Chapter 5 Consultation and Revision	AZ, AS	3
BB	Rechecking of Chapter 5 Revisions	BA	2
BC	Chapter 5 Final Submission	BB	1
BD	Manuscript Synthesis	AX, BC, AY, I, M, R, AO	4
BE	Final Oral Defense Presentation	BD	5
BF	Deployment and Handover	BE	10

PERT Chart

The PERT chart serves as a strategic project management tool that maps the complex dependencies between dataset operations and AI model development to determine the most efficient timeline for completion by November 2026. By organizing the research into five structured phases, the chart utilizes a node-based system to calculate Early Start and Late Finish dates, identifying the critical path

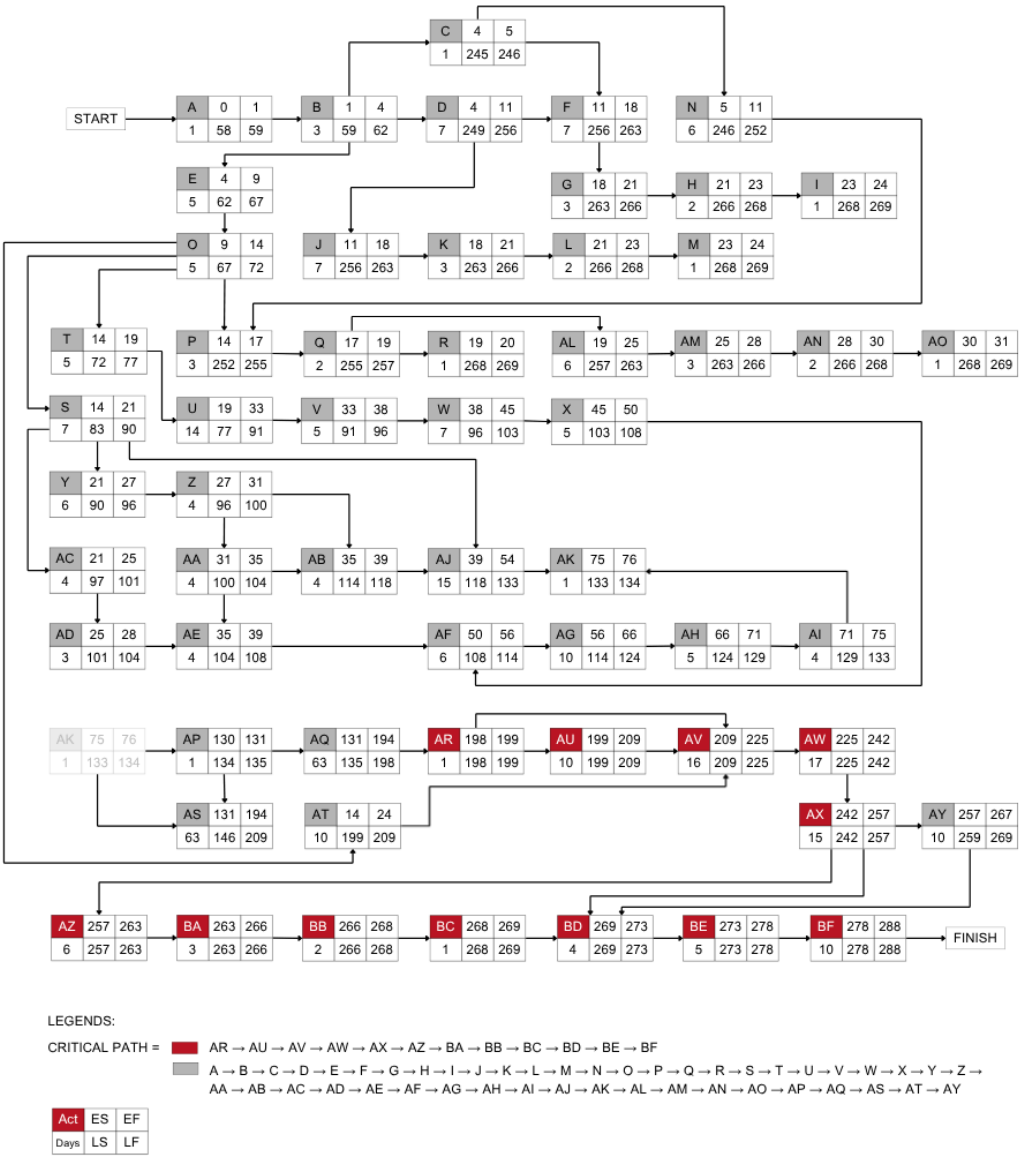


Figure 4.2. *PERT Chart*

Designing Phase

After the initial planning, this phase focused on establishing the technical framework of the application by defining the system architecture and data flow structures required to handle real-time image input, on-device multi-class inference, Grad-CAM generation, and condition-specific first-response guidance retrieval for fully offline output delivery. This stage involved the development of context diagrams and data flow diagrams to map the movement of information between users and the screening pipeline. The logic for transforming raw camera images into multi-class classification results, Grad-CAM heatmaps, and condition-matched first-response recommendations was specified through the alignment of the neural network inference pipeline and bundled knowledge base with the Flutter application's offline-first delivery architecture. These structural blueprints served as the technical foundation for the development and integration of all core system components.

Context Diagram

The Context Diagram defines the operational boundary of Scanine by mapping the data exchanged between the system and its sole external actor, the User. The interaction begins when the User submits a Capture Canine Lesion Images request, prompting the system to process the input and return one of two outcomes. If the submitted image fails the system's internal quality check, Scanine responds with Image Quality Feedback, prompting the user to recapture the photo before any classification is attempted. Once an image clears that check, the system returns three outputs together: a Classification Result identifying the detected condition (Ringworm, Hotspot, Flea Allergy Dermatitis, or Dermatologically Normal), a Grad-CAM Heatmap Overlay that visually pinpoints the image regions the model relied on to

reach that decision, and First-Response Guidance, condition-matched interim care instructions the user can act on immediately. Because this diagram sits at the highest level of abstraction, it deliberately leaves out internal processing detail; its purpose is only to establish that Scanine functions as a single, self-contained processing boundary the user interacts with directly, without needing to know what happens inside.

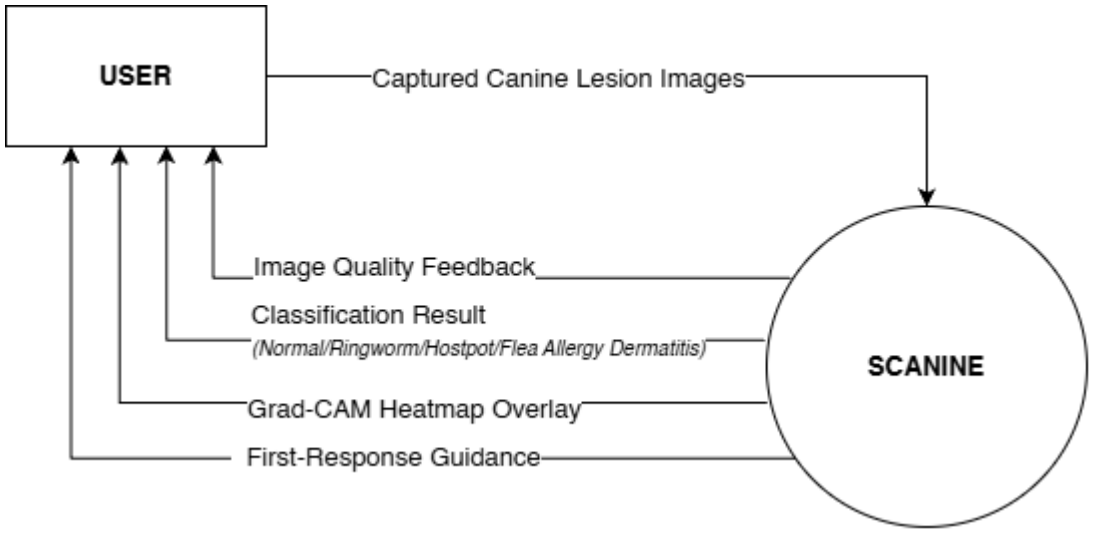


Figure 4.3. *Context Diagram of the Proposed System*

The traditional process of assessing canine dermatological conditions in community and field settings follows a multi-phase, manual workflow heavily reliant on human judgment and institutional coordination. For dermatophytosis, confirmation requires Wood's lamp examination, fungal culture, or microscopic analysis of hair shafts conducted under laboratory conditions, a process that unfolds over 7 to 14 days from specimen collection to diagnosis. Hotspot cases, while clinically faster to progress, depend entirely on the subjective recognition of an untrained observer and are frequently escalated only after the lesion has spread through continued self-trauma. Flea allergy dermatitis similarly goes unmanaged until pruritic signs become severe, as

intradermal allergen testing and therapeutic elimination trials are inaccessible outside a clinical setting. Across all three conditions, the initial point of contact follows the same constrained sequence: a skin condition is visually spotted and evaluated without diagnostic tools or standardized protocol; unsuspicious cases are released with a critical screening gap for potential spread; and suspicious cases trigger a report to the Animal Welfare Organization (AWO), where delays of hours to days are common before a subjective, unstandardized assessment can be conducted. For cases that do proceed to clinical evaluation, the animal must be physically captured, transported, and examined at a veterinary facility before any confirmed output is produced. The lengthy timeline, multiple points of potential misdiagnosis, and complete absence of standardized field-level screening tools across all phases underscore the significant limitations of this traditional approach and highlight the pressing need for a more reliable, technology-assisted multi-condition screening solution.

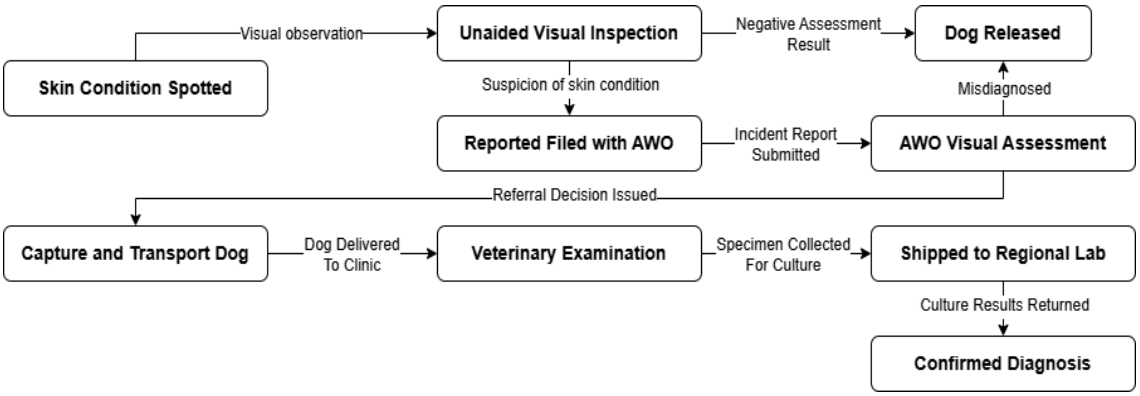


Figure 4.4. *Manual Screening Process for Canine Skin Diseases*

Data Flow Diagrams

The Level 1 Data Flow Diagram decomposes the system into four sequential processes, each supported by its own dedicated data store, showing how a single captured image is progressively transformed into a complete triage output. The pipeline opens at Process 1.0, On-Device Quality Filtering, which receives the Captured Canine Lesion Images from the User and checks it against D1: On-Device Quality Filter Parameters (Hive Local DB), a local store holding the blur and lighting thresholds an image must clear. Images that fail are rejected immediately, with Image Quality Feedback returned to the user; images that pass move forward as a Validated Image Frame into Process 2.0, Multi-Class Inference, which retrieves the trained weights from D2: MobileNetV3 + CoordAttention Weights (TFLite Model) to run the actual classification. This produces the Classification Result delivered to the user, while the frame and its extracted feature data are passed onward to Process 3.0, Visual Explainability (Grad-CAM), which draws on D3: Gradient Activation Store (RAM Cache), a temporary in-memory store holding the gradient data needed to compute the heatmap, to generate the Grad-CAM Heatmap Overlay returned to the user. Only the Detected Category is forwarded from there to Process 4.0, First-Response Guidance Retrieval, which uses that category to query D4: Static Vet-Reviewed Knowledge Base (Hive Local DB) and retrieve the matching guidance protocol, delivered to the user as First-Response Guidance. Read end to end, the diagram shows a clean linear handoff: each process consumes only what the previous one produced, and each store serves exactly one process, keeping classification, explainability, and guidance cleanly separated even though all three operate on the same underlying image.

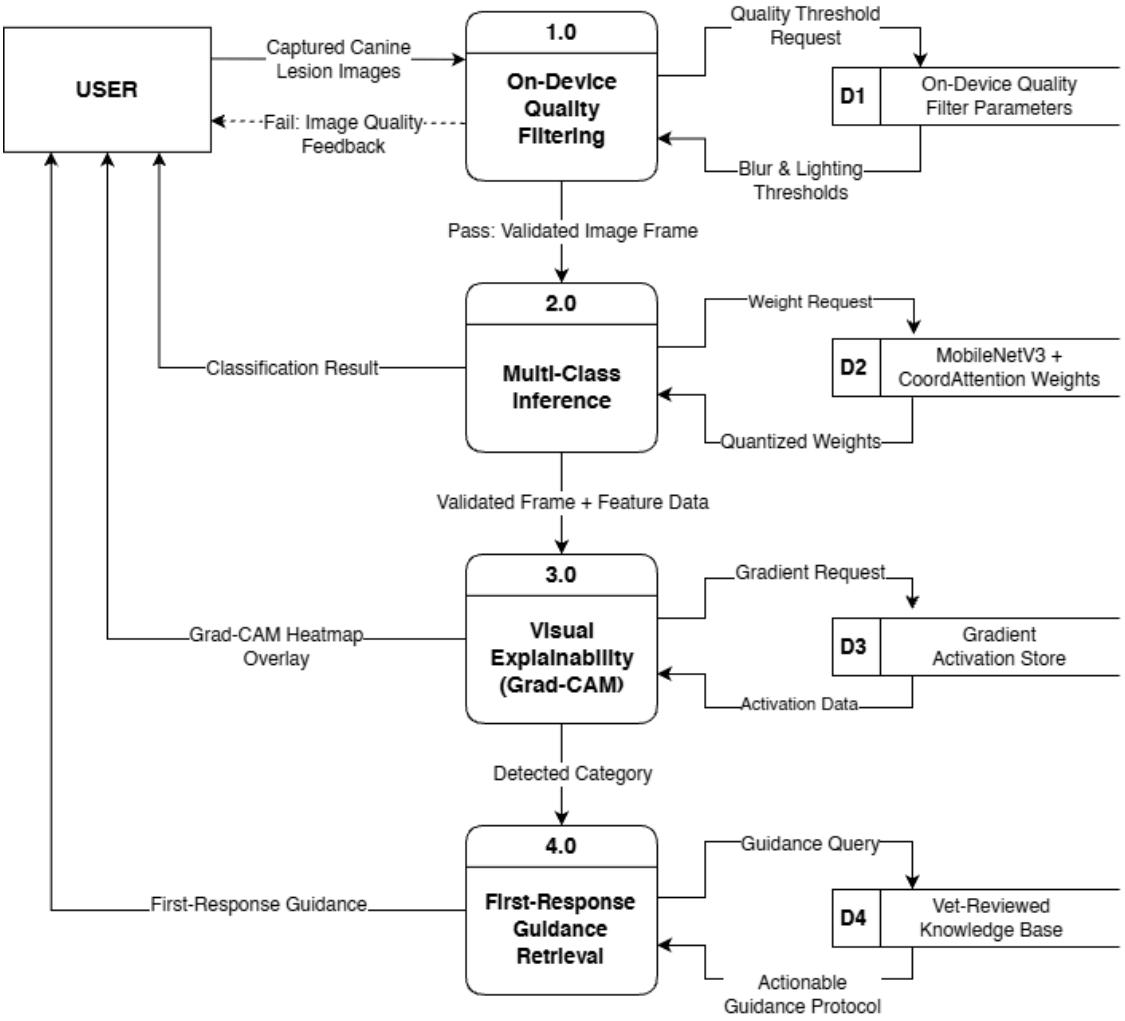


Figure 4.5. *Level 1 Data Flow Diagram of the Proposed System*

Breaking down Process 1.0 into its Level 2 sub-processes, the following diagram details the internal flow of On-Device Quality Filtering. As shown in Figure 4.6, the process begins at sub-process 1.1, Frame Ingestion, where the application receives the image supplied by the user and converts it into a Raw Image Array. This array is then routed along two parallel evaluation paths. In 1.2, Lighting Assessment, the system sends an Exposure Threshold Request to D1: On-Device Quality Filter Parameters (Hive Local DB) and compares the image's luminance against the retrieved Exposure Threshold to produce a Lighting Metric Score. Simultaneously, in 1.3, Blur

Detection, the same Raw Image Array is evaluated for sharpness by requesting the Laplacian Variance Thresholds from the same D1 store and computing a Blur Metric Score. Both scores converge at 1.4, Quality Dispatch, where a final pass-or-reject decision is made: images that meet both criteria are forwarded as a Validated Image Frame to Process 2.0, while those that do not trigger Image Quality Feedback returned to the user, prompting recapture under better conditions. This dual-check filtering layer is essential to the system's diagnostic integrity, as it prevents substandard input from reaching the classification model under the variable conditions of real-world field deployment.

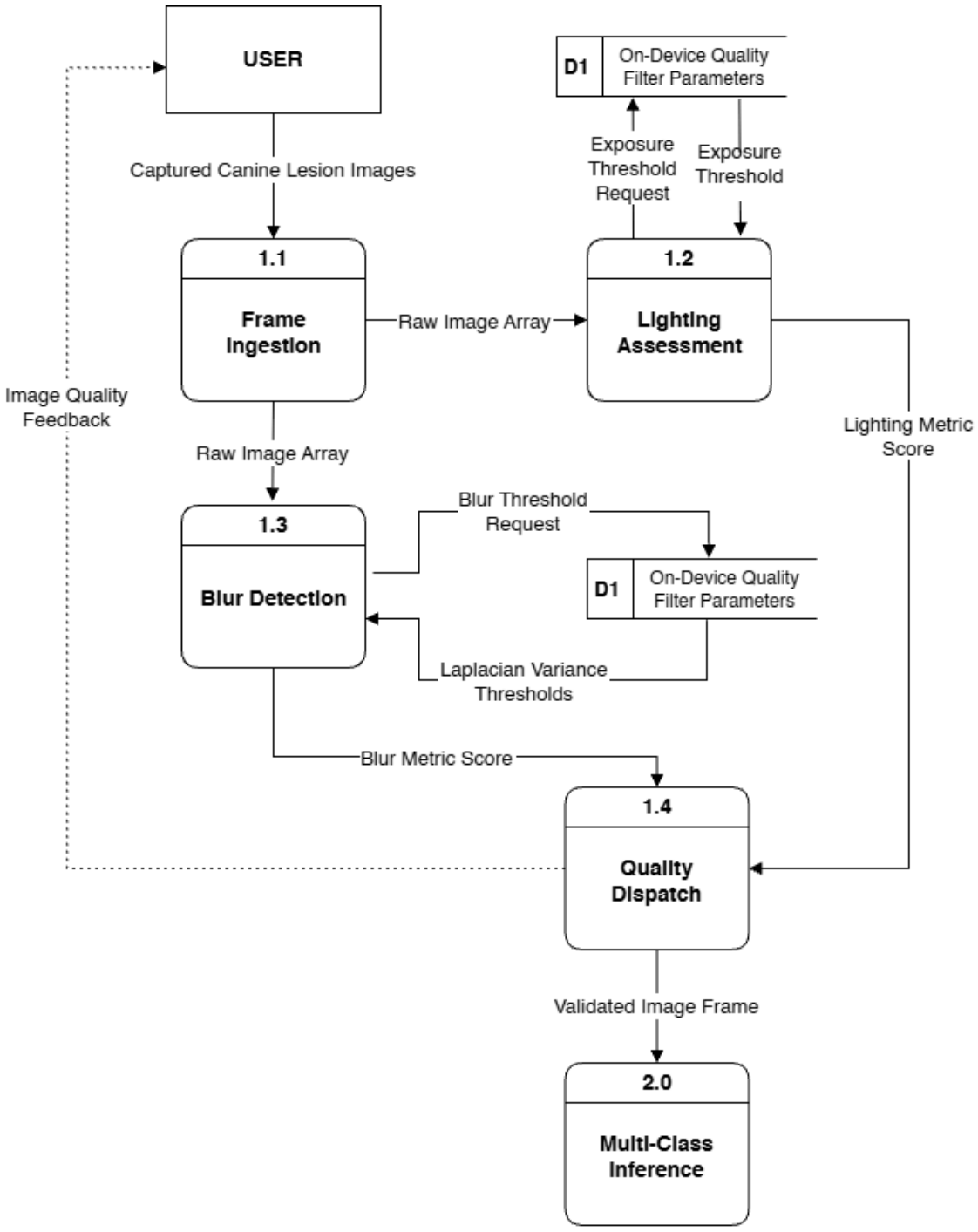


Figure 4.6. *Level 2 DFD: Process 1.0 (On-Device Quality Filtering)*

Upon successful validation, the data stream transitions into Process 2.0, Multi-Class Inference, detailed in Figure 4.7. Operations begin at 2.1, Input Tensor

Preprocessing, where the validated image is converted into a Preprocessed Tensor Array suitable for the neural network. This tensor moves to 2.2, Model Weight Loading, which sends a Weight Request to D2: MobileNetV3 + CoordAttention Weights (TFLite Model), receives the Quantized Weights in return, and assembles the Prepared Tensor & Model Parameters needed to run the network. At 2.3, Forward Pass Execution, this package is processed through the model backbone to produce Raw Logits & Spatial Activation Maps, the unprocessed prediction scores together with the intermediate feature maps the network relied on to generate them. The pipeline concludes at 2.4, Prediction Post-Processing, which converts these raw logits into the final Classification Result delivered to the user, while forwarding the Validated Frame + Feature Data, the same spatial activation maps produced during the forward pass, to the Visual Explainability module.

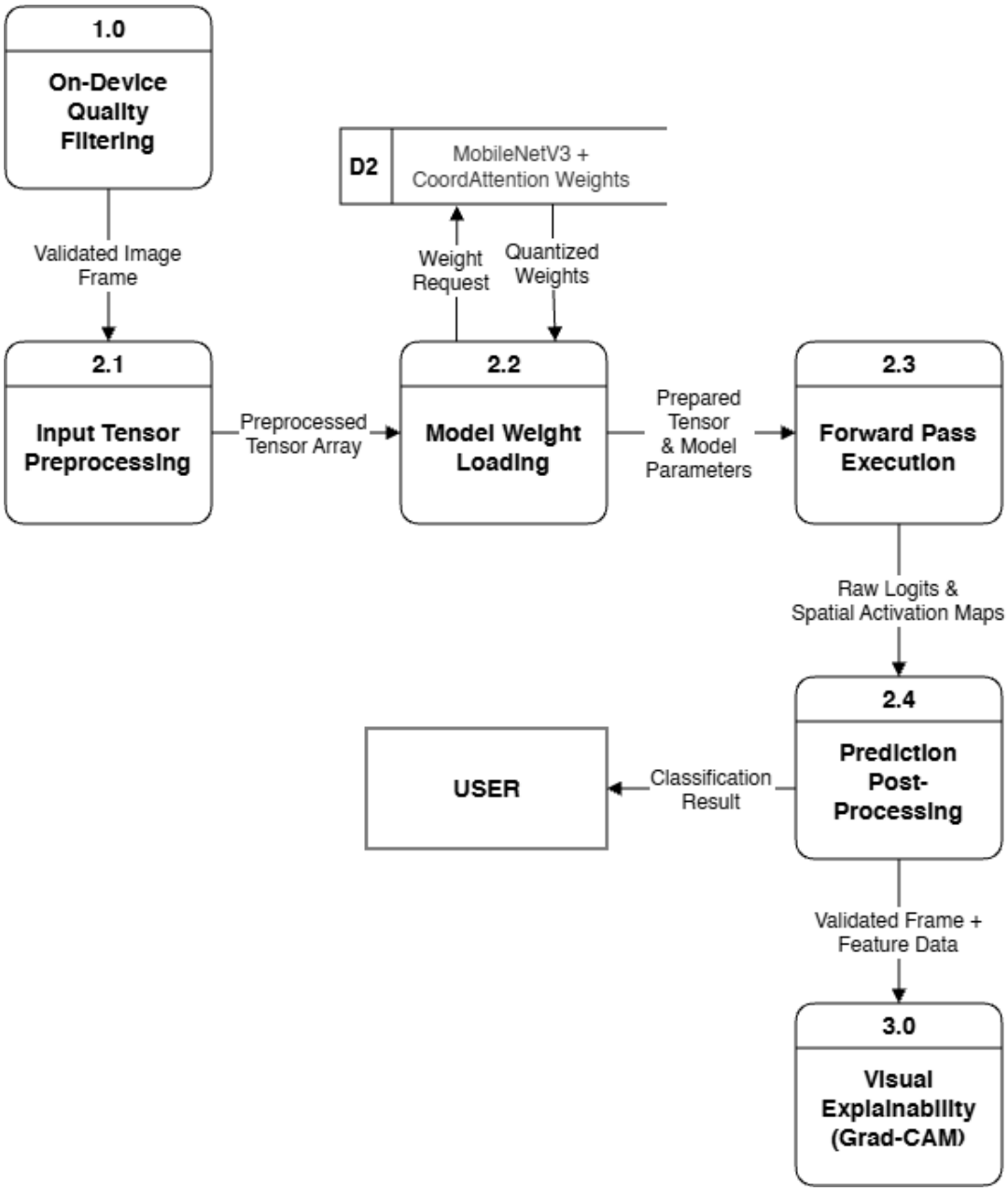


Figure 4.7. *Level 2 DFD of Process 2.0: Multi-Class Inference*

To bridge the gap between inference and clinical trust, Process 3.0 provides visual explainability through the Grad-CAM technique, as illustrated in Figure 4.8. This module initiates at 3.1, Feature Mapping, which receives the Validated Frame and Feature Data from the inference engine and produces Spatial Feature Tensors. The

process flow progresses to 3.2, Gradient Computation, where the system sends a Gradient Request to D3: Gradient Activation Store (RAM Cache) and receives Activation Data to compute Gradient Importance Weights. The pipeline concludes at 3.3, Visualization Engine, where the system renders the Grad-CAM Heatmap Overlay and delivers it to the user, while simultaneously forwarding the Detected Category to Process 4.0 to initiate condition-specific guidance retrieval.

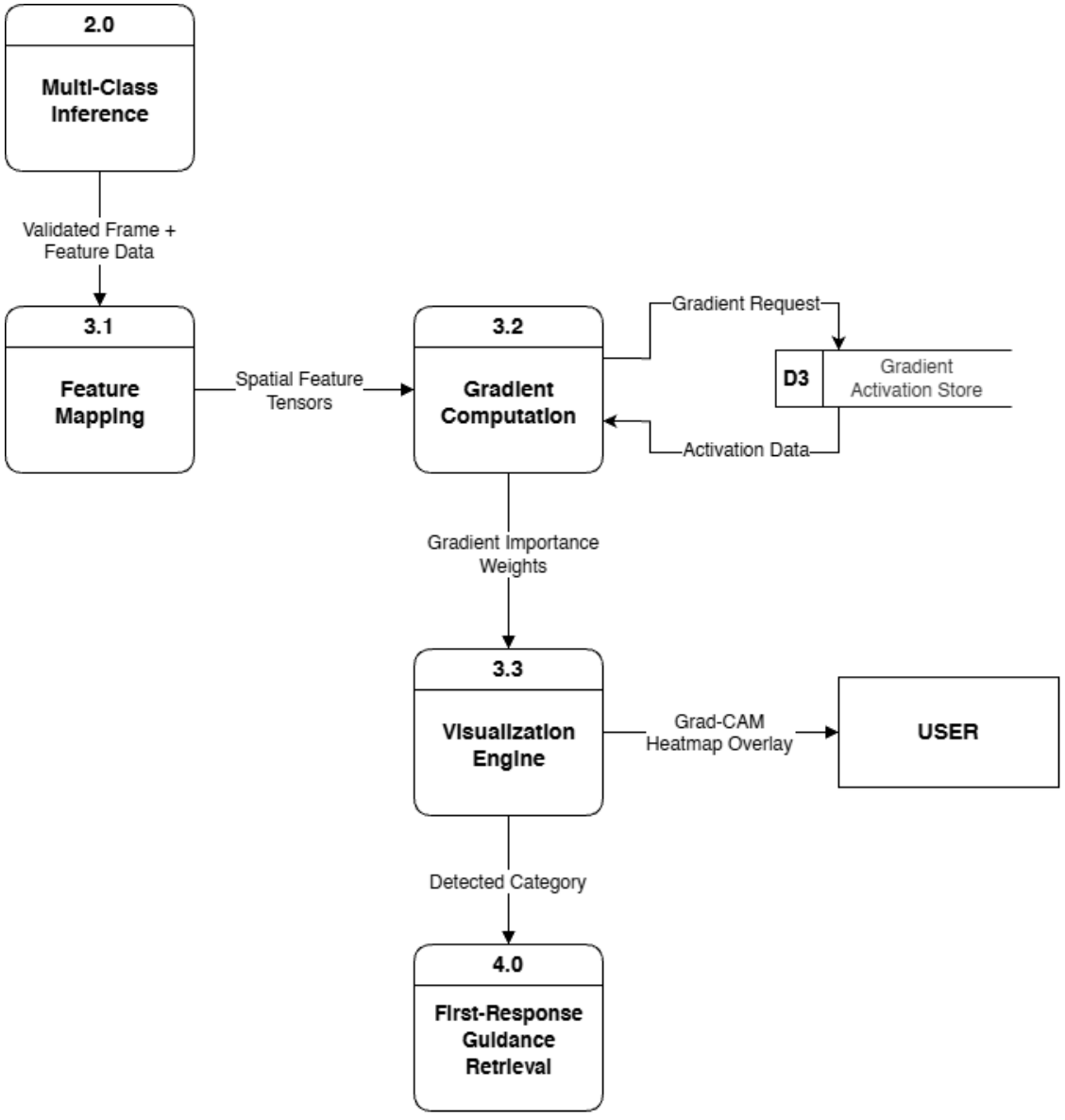


Figure 4.8. *Level 2 DFD: Process 3.0 (Visual Explainability (Grad-CAM))*

The final stage of the analytical routine is Process 4.0, First-Response Guidance Retrieval, which receives the Detected Category produced by the Visualization Engine in the preceding explainability step, as shown in Figure 4.9. Sub-process 4.1, Category Indexing, maps the detected condition to a Database Key corresponding to the appropriate entry in the knowledge base. At 4.2, Protocol Fetching, the system issues a Guidance Query to D4: Static Vet-Reviewed Knowledge Base (Hive Local DB) and retrieves the corresponding Raw Protocol Data. The pipeline culminates at 4.3, Guidance Formulation, where the retrieved protocol is formatted and delivered as First-Response Guidance to the user, completing the transformation of a raw classification result into a condition-matched, actionable interim care recommendation ready for field use without requiring network access.

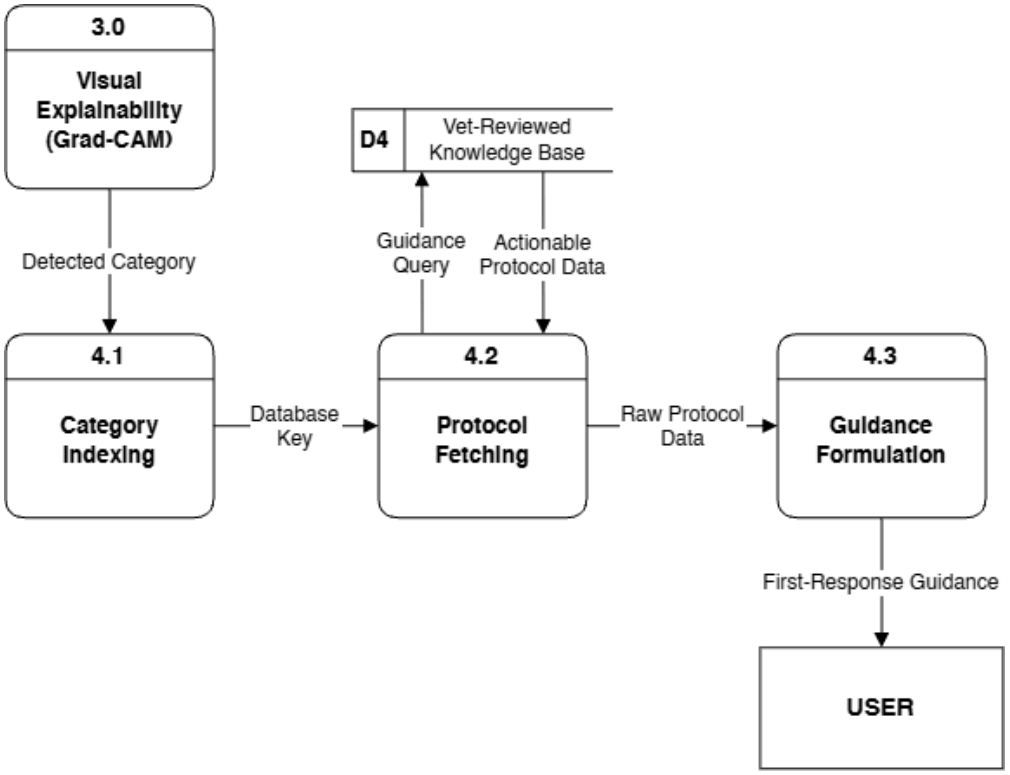


Figure 4.9. *Level 2 DFD: Process 4.0 (First-Response Guidance Retrieval)*

Development Phase

Following the completion of the design blueprints, the development phase operationalized the system's technical architecture into a functional screening application. This phase centered on the training, compression, and on-device deployment of the multi-class classification model within the Flutter application, the implementation of the Grad-CAM module and first-response guidance retrieval pipeline, and the construction of the offline-first mobile application framework. Underlying each of these components is a shared image processing pipeline responsible for capturing, preprocessing, and transforming raw visual input into the structured data from which all subsequent classification, interpretability, and guidance outputs are derived.

Image Processing Pipeline

Figure 4.10 presents the complete processing architecture of Scanine, organized as seven sequential stages that transform a raw camera capture into a classified condition, a visual explanation of that classification, and a corresponding first-response recommendation. The architecture is governed throughout by a single non-negotiable constraint: every stage executes locally on the user's device, and no stage issues a network request at any point. This constraint is not incidental to the design but generative of it. It determines the selection of a mobile-optimized convolutional backbone, the decision to convert the trained model to TensorFlow Lite for on-device inference, the treatment of first-response guidance as a bundled static asset rather than a served resource, and — as discussed below — the manner in which gradient-based explanations are computed. The pipeline is presented here as the

deployed runtime path; model training, cross-validation, and quantization occur offline on a workstation and are documented separately.

The pipeline begins at the image acquisition stage, which captures five sequential frames rather than a single photograph. This decision reflects the operating conditions the system is designed for: non-expert users — pet owners, animal welfare volunteers, and community handlers — photographing uncooperative and frequently free-roaming animals in uncontrolled outdoor lighting, often through dense fur. Under such conditions capture quality, rather than model capacity, constitutes the dominant source of classification error. The system therefore addresses image quality at its source through two sequential on-device gates. Blur is assessed by the variance of the Laplacian, a computationally inexpensive second-derivative operator whose response falls sharply as an image loses high-frequency edge detail; captures falling below the blur threshold are rejected as insufficiently sharp. Frames passing this gate are then evaluated for exposure by mean pixel intensity, with captures outside an acceptable luminance band rejected as underexposed. Substandard frames are identified and excluded *before* any inference is attempted, so that the classifier is never asked to interpret an image from which reliable features cannot be extracted.

Validated frames proceed to preprocessing, where four deterministic transformations condition the raw capture into the tensor format expected by the network. The frame is first resized to 224 × 224 pixels by bilinear interpolation, then cast from unsigned 8-bit integer to 32-bit floating point, normalized from the range [0, 255] to [−1, +1] by the transformation $x' = (x - 127.5) / 127.5$, and finally expanded along a leading batch axis to yield a tensor of shape [1, 224, 224, 3]. The ordering of these operations is deliberate: resizing precedes normalization so that interpolation is

performed in the original intensity space and so that no arithmetic is expended on pixels subsequently discarded. Of greater consequence, this preprocessing chain is defined once and shared identically with the training pipeline. Divergence between training-time and inference-time preprocessing — in interpolation method, channel ordering, or normalization constants — produces accuracy degradation that manifests as an apparent model defect while originating in the data path, and is correspondingly difficult to diagnose from the application layer.

The preprocessed tensor enters the feature representation stage, implemented as a MobileNetV3-Small convolutional backbone initialized from ImageNet pre-trained weights. The backbone employs depthwise separable convolutions, which factorize standard convolution into per-channel spatial filtering followed by pointwise channel mixing, substantially reducing parameter count and multiply-accumulate operations relative to conventional convolution, and hard-swish activations, a piecewise-linear approximation of the swish function selected for its efficiency under fixed-point arithmetic. Coordinate Attention instead applies average pooling along the height and width axes independently, concatenates the resulting one-dimensional descriptors, processes them through a shared 1×1 convolution with batch normalization and non-linearity, and splits the result into two direction-aware attention maps that are applied multiplicatively to the input. Positional information is thereby retained within the attention mechanism. For dermatological screening — a task in which lesion extent, morphology, and anatomical location carry diagnostic significance — this positional sensitivity constitutes the substantive justification for the substitution. The stage yields final convolutional feature maps $A[k]$ of dimension $7 \times 7 \times 576$, which serve both the classification head and, subsequently, the generation of visual explanations.

Classification and decision converts these feature maps into a single reported outcome. Global average pooling reduces the spatial dimensions of A[k] to a 576-element descriptor, which a softmax classification head maps to a probability distribution over four classes: dermatophytosis (ringworm), hotspot, flea allergy dermatitis, and normal skin. Because five frames are captured per screening event, this procedure yields five independent probability distributions, which are consolidated by temporal smoothing through plurality vote across frame-level predictions or, equivalently, selection of the class with the highest mean softmax response. Beyond stabilizing the prediction against transient capture artifacts, this multi-frame consolidation functions as a form of test-time augmentation, an effective technique where training data are limited. The consensus prediction is then subjected to a confidence gate: where the maximum calibrated probability falls below a predetermined threshold, the system declines to assert a class and instead returns an inconclusive result directing the user to seek veterinary consultation. This abstention path addresses a specific and foreseeable failure mode. The classifier recognizes three conditions, while the population of canine dermatological presentations is considerably larger; an animal presenting with mange, pyoderma, or any condition outside the training distribution would otherwise be assigned to one of four classes and issued a confidently-worded recommendation. Permitting the system to withhold judgment converts a silent misclassification into an honest referral.

The explainability stage generates a Grad-CAM heatmap localizing the image regions that most influenced the prediction. Frame selection first identifies which of the five captures the explanation should describe, choosing the highest-confidence frame whose individual prediction agrees with the temporal consensus. The class activation map for that frame is obtained as a second output of the same forward pass

that produced the logits, rather than through a separate computation. This arrangement follows from the deployment constraint. Grad-CAM in its canonical formulation requires the gradient of the target class logit with respect to the final convolutional activations, $\partial y_c / \partial A[k]$, and TensorFlow Lite is a forward-only interpreter that exposes no gradient computation at runtime. The gradient is therefore evaluated within the TensorFlow graph *prior* to model conversion, such that the converter traces the backward pass and emits it as ordinary forward operations. Because the gradient is taken only with respect to the terminal convolutional layer, this traced path spans the classification head alone rather than the full backbone, and the resulting computational overhead is marginal. The raw 7×7 activation map is bilinearly upsampled to 224×224, normalized and colorized through a jet colormap, and composited onto the selected frame. Consistent with the acknowledged limitations of the technique, the resulting overlay is presented to users as a coarse indication of model attention and not as a delineation of lesion boundaries.

Condition-specific guidance translates the validated classification into actionable instruction through a first-response matrix engine. Guidance content is drawn from a vet-reviewed content store: a static, pre-authored corpus reviewed by a veterinary professional for clinical safety and bundled with the application at build time. This constitutes a deliberate trade-off favouring reviewability over adaptiveness — because the guidance is fixed and enumerable rather than generated, every statement the system can make to a user has been examined and approved in advance, which is the basis of the system's clinical-safety claim. Class probability routing directs the validated condition target to the corresponding entry, a rule-based recommendation engine performs deterministic lookup without any generative component, and protocol formatting applies urgency scaling appropriate to the

identified condition. Ringworm receives particular treatment at this stage: as a zoonotic condition transmissible to the humans handling affected animals, and a principal element of the rationale for siting field evaluation in Tacloban City, a ringworm outcome attaches a human-safety advisory in addition to animal-care instruction.

The final system output presents four artifacts concurrently: the predicted class with its associated confidence percentage, the underlying five-frame vote result, the Grad-CAM heatmap overlay, and the condition-specific first-response guidance. A persistent screening disclaimer accompanies all results, stating that the output constitutes preliminary screening only, does not represent a veterinary diagnosis, and does not replace professional consultation. Situating this statement within the output layer, rather than confining it to accompanying documentation, renders the limitation a property of the artifact itself. Taken as a whole, the architecture reflects three converging design priorities: offline operability, which shapes every implementation decision from backbone selection to the placement of gradient computation; robustness to the capture conditions non-expert users will realistically encounter, addressed through quality gating and multi-frame consensus; and epistemic restraint appropriate to a screening instrument, expressed through calibrated abstention, coarse rather than precise visual explanation, and pre-reviewed rather than generated advice.

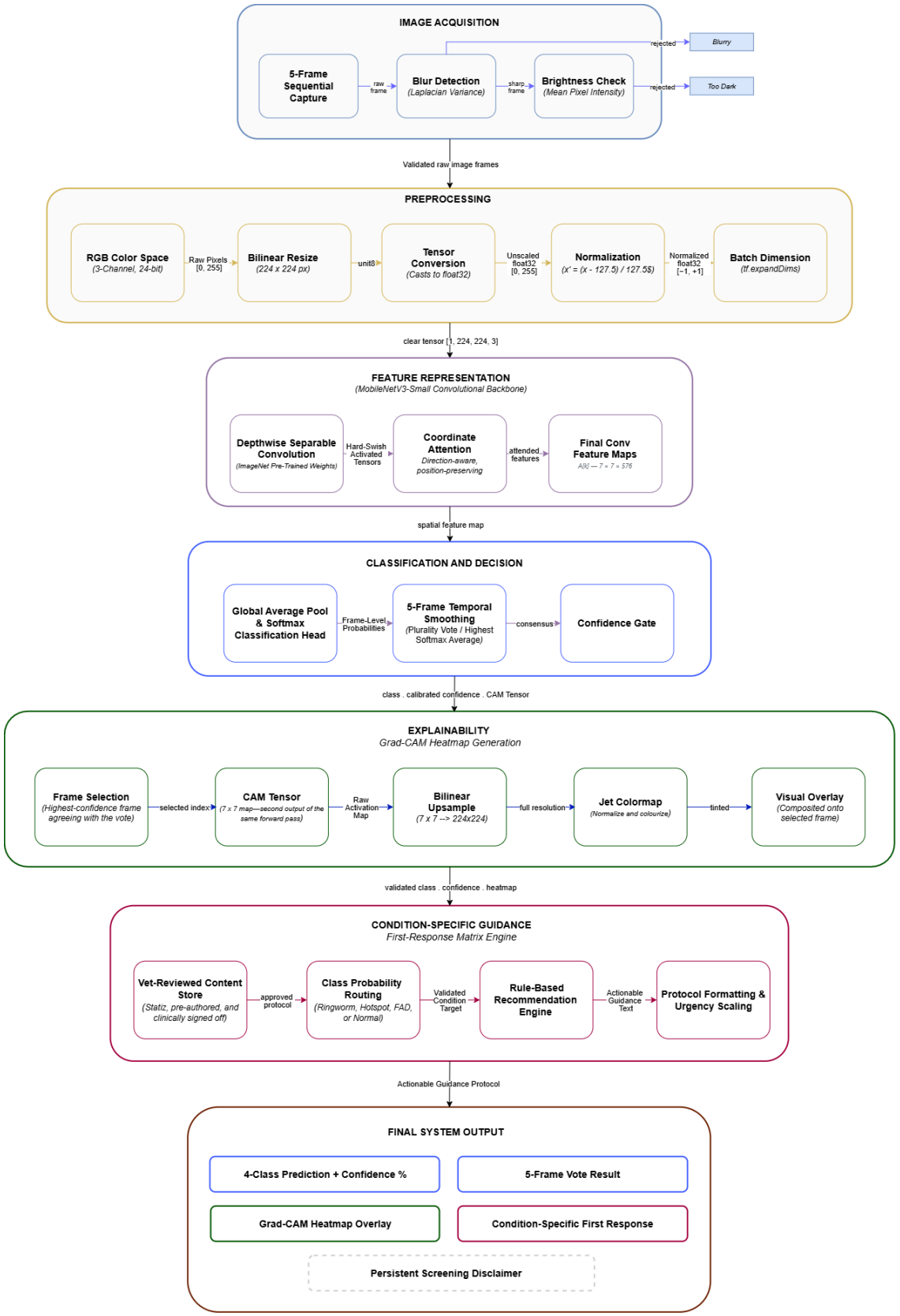


Figure 4.10. *Image Processing Pipeline*

Testing Phase

White-Box Testing

The testing phase utilized automated unit and integration testing at the module level to evaluate the internal logic of the application's core processing components. Automated test suites verified that the image quality filter correctly applied the Laplacian variance and brightness thresholds and routed images to the appropriate downstream process based on their quality score. Within the inference pipeline, specific test cases validated that the TensorFlow Lite runtime correctly loaded the quantized model weights bundled with the application and produced multi-class classification outputs consistent with the model's validated behavior on a held-out test set. The first-response guidance retrieval module was tested against all classification categories to confirm that the correct condition-specific protocol was retrieved from the knowledge base for each detected class and that the guidance output was correctly formatted and delivered to the user interface.

Following unit verification, the system underwent integration testing to assess end-to-end functionality according to the ISO/IEC 25010 Functional Correctness and Performance Efficiency standards. This stage verified the complete screening pipeline from camera image capture through quality filtering, multi-class inference, Grad-CAM generation, guidance retrieval, and report delivery. Tests confirmed that the Grad-CAM heatmap overlay was spatially aligned with the input image for each condition class and that the condition-specific first-response guidance was consistently matched to the corresponding classification output across repeated runs of the same input. Offline functionality was verified by disabling network access after installation

and confirming that all screening features, including Hive-based report logging, remained fully operational across subsequent sessions.

Black-Box Testing

User acceptance testing followed, focusing on functional utility rather than internal code. Veterinary professionals and community responders assessed whether the application effectively translates the classification result, Grad-CAM heatmap, and first-response guidance into a coherent and actionable screening output. The goal was to confirm that the interface is sufficiently intuitive for non-specialist field users and that the triage outputs support practical decision-making for canine dermatological screening in stray dog populations. To quantify this functional utility, the evaluation utilized a 5-point Likert scale.

Table 4.6. *5-Point Likert Scale*

Scale	Qualitative Descriptor	Interpretation
5	Strongly Agree	The system functions optimally; outputs are clearly interpretable and the interface is intuitive for field use.
4	Agree	The system meets functional requirements with minor, non-disruptive usability issues.
3	Neutral	The system is functional, but improvements in interface design or output presentation are necessary.
2	Disagree	The system fails to present outputs clearly or the interface hinders task completion in the field.
1	Strongly Disagree	The system is non-functional or the outputs do not support real-world triage decision-making.

Participant Selection

The study employs Purposive Sampling to select evaluators with the domain knowledge required to assess the system's screening outputs and interface utility. This non-probability sampling method ensures that participants possess the necessary expertise to provide high-validity feedback on both the technical performance and practical usability of the application.

Table 4.7. *Participant Inclusion and Exclusion Criteria*

Category	Description
Inclusion	Licensed veterinary professionals or trained community animal welfare responders actively involved in canine health monitoring or stray animal management; minimum of one year of experience in field-based animal health screening or community animal care; willingness to use digital screening tools in field settings..
Exclusion	No experience in animal health monitoring or community animal welfare activities; commercial conflicts of interest regarding competing animal health screening tools; incomplete participation in the mandatory system orientation session; inability to provide informed consent or complete the evaluation protocol.

Ethical Considerations

All evaluators and users involved in the testing phase are provided with clear and complete information about the purpose, procedures, and scope of the study. They are informed of their roles and how their participation contributes to the evaluation of

the screening application. Consent is obtained prior to any testing activity, ensuring that involvement is fully voluntary and based on informed understanding.

All collected data, including system outputs and evaluation feedback, are treated with strict confidentiality. Identities are anonymized to protect privacy, and access to evaluation information is limited to authorized individuals involved in the study. These measures ensure that data remain secure throughout the testing process.

Participation is entirely voluntary, with no form of coercion or pressure. Individuals are free to decline involvement or withdraw at any stage without negative consequences. This upholds respect for individual autonomy and ensures the integrity of the evaluation process.

All data generated during testing are handled responsibly and used strictly for research and validation purposes. Screening images and evaluation records are protected from unauthorized access or disclosure, and proper data management practices are followed throughout, consistent with the requirements of the Data Privacy Act of 2012.

Implementation Phase

In this phase, Scanine is deployed within controlled field environments, connecting the image capture pipeline, on-device inference engine, Grad-CAM module, and first-response guidance retrieval system into a unified and fully offline screening workflow. Developers finalize the classification model and knowledge base bundled with the application package, validate offline inference execution, confirm Hive-based report logging, and verify condition-specific guidance output consistency across target device configurations, preparing the system for real-world use by

community responders. This stage bridges the gap between the technical design and its practical application, ensuring that the application operates reliably under field conditions, delivers interpretable and condition-grounded screening outputs, and is ready for stakeholder evaluation and veterinary expert assessment.

Use Case

The Use Case Diagram illustrates the functional interactions between the User and Veterinary Professional actors, as shown in Figure 4.11. The User initiates the workflow via Capture or upload image, which serves as the primary input for the Run AI Screening use case. This process allows the User to subsequently View Grad-CAM Heatmap and View First-Response Guidance to review the spatial activation evidence and condition-matched interim care recommendations for the detected presentation. Finally, the User can perform the Refer to Veterinarian use case, which facilitates the transition from preliminary screening to professional assessment by involving the Veterinary Professional. This configuration ensures that the User has access to both localized heatmaps and condition-specific guidance as structured triage outputs before deciding to escalate a case for professional veterinary verification.

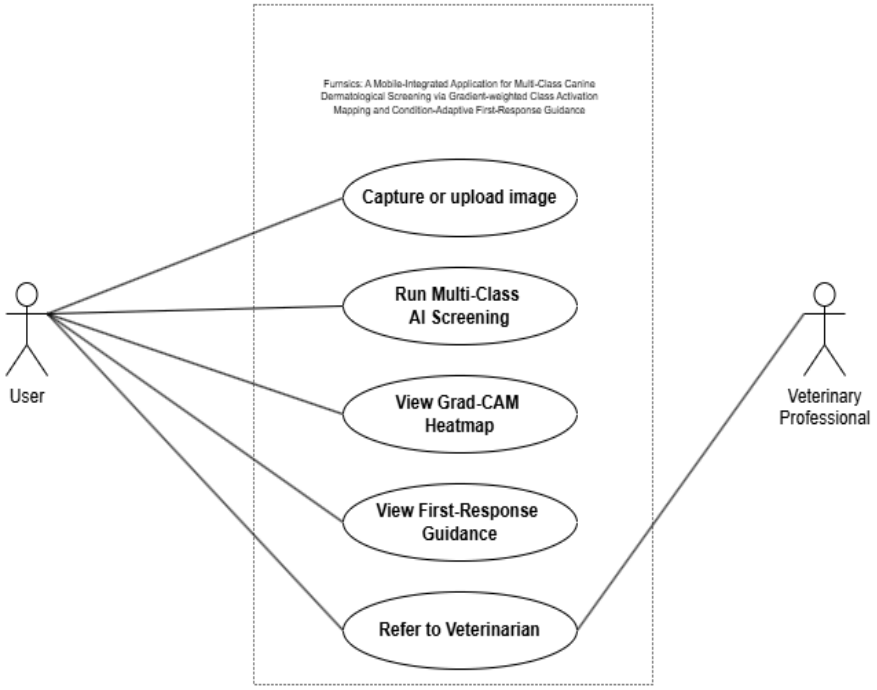


Figure 4.11. *Use Case Diagram*

Implementation Plan

Project Title

Scanine: A Mobile-Integrated Application for Multi-Class Canine Dermatological Screening via Grad-CAM and Condition-Adaptive First-Response Guidance

Project Description

This project addresses a critical gap in community-level animal health surveillance by developing a mobile-integrated screening tool that enables non-specialist community responders to perform standardized multi-condition canine dermatological triage using a standard smartphone. The platform operates as an offline

mobile application built in Flutter, executing all classification, explainability, and guidance retrieval computations directly on the user's device without requiring an internet connection during operation. Unlike general-purpose canine skin disease classifiers, Scanine is purpose-built for the simultaneous detection of three clinically and epidemiologically distinct conditions: dermatophytosis, hotspot, and flea allergy dermatitis, with a classification model calibrated for high sensitivity to minimize missed detections across all condition classes. The system prioritizes interpretability and triage utility by generating Grad-CAM activation heatmaps that spatially localize pathological skin texture for each detected condition, and delivers condition-matched first-response recommendations retrieved from a knowledge base bundled with the application, providing actionable interim guidance calibrated to each condition, including isolation protocols for dermatophytosis-suspected animals, environmental hygiene measures for flea allergy dermatitis cases, and self-trauma prevention for hotspot presentations, without requiring an additional machine learning model or server dependency.

Objectives

The primary goal of the implementation phase is to validate that the integrated system performs its full multi-class screening workflow reliably under field-representative conditions and produces outputs that are both technically consistent and practically interpretable by its intended users. The system is designed to accept standard smartphone camera input and deliver, within a single screening session, a multi-class classification result identifying the detected condition, a Grad-CAM heatmap overlay spatially localizing the pathological region for that condition, and condition-specific first-response guidance retrieved from the bundled

knowledge base, together constituting a structured triage output. By rejecting images that fail the internal quality filter and retrieving condition-indexed guidance from the offline knowledge base, the implementation validates that the system maintains output reliability and condition specificity even under the variable lighting and motion conditions of outdoor canine screenings. Ultimately, the implementation aims to demonstrate that Scanine is a technically viable and practically usable triage aid for community responders conducting multi-condition canine dermatological surveillance in Tacloban City, providing a standardized, non-invasive alternative to subjective visual inspection without requiring laboratory infrastructure or specialist personnel.

Duration

The technical development and deployment of the application are scheduled across five overlapping phases, with full operational turnover targeted for November 2026. Using the Sashimi model, the timeline allows for the simultaneous development of the multi-class machine learning pipeline and the Flutter application frontend, with Grad-CAM integration and first-response guidance module development conducted in parallel with interface refinement. This duration encompasses the entire project lifecycle, including dataset curation across three condition classes, model training and quantization, knowledge base construction, system integration, stakeholder testing, and a dedicated phase for community responder orientation to ensure the application is effectively deployed in its intended field context.

Table 4.8. *Implementation Plan (Tabular)*

Module	Specific Objectives	Hardware Requirements	Software Requirements	Duration	Person Involved	Estimated Budget	Risk Assessment & Mitigation Strategies	Evaluation & Monitoring
Multi-Class Dataset Acquisition and Preparation Module	Collect, validate, annotate, and preprocess four-class canine dermatological image datasets (dermatophytosis, hotspot, flea allergy dermatitis, dermatologically normal) for AI model training and evaluation	Personal computer (minimum 8GB RAM), local storage, smartphone	Python, OpenCV, LabelImg/Roboflow, Pandas, NumPy	5 weeks	Project Proponents	N/A	Risk: Incomplete or low-quality image datasets across the four condition classes. Mitigation: Apply image quality screening, augmentation, and per-class validation procedures.	Monitor dataset consistency, annotation accuracy, and class balance distribution across all four classes.
Multi-Class Classification Model Development Module	Develop and train a MobileNetV3-Small backbone enhanced with Coordinate Attention for four-class classification of canine dermatological conditions, followed by 8-bit quantization for on-device deployment	Personal computer or workstation (16GB RAM recommended)	Python, TensorFlow, TensorFlow Lite, Scikit-learn	4 weeks	Project Proponents	₱2,500 (model training and testing resources)	Risk: Model overfitting or reduced sensitivity across the four condition classes. Mitigation: Apply cross-validation, data augmentation, and hyperparameter tuning, with per-class validation to catch minority-	Evaluate validation accuracy, precision, recall, and Area Under the Curve (AUC) per condition class.

Module	Specific Objectives	Hardware Requirements	Software Requirements	Duration	Person Involved	Estimated Budget	Risk Assessment & Mitigation Strategies	Evaluation & Monitoring
							class underperformance.	
Grad-CAM Explainability Module	Implement Gradient-weighted Class Activation Mapping (Grad-CAM) to generate localized activation heatmaps identifying the image regions driving each classification decision	Personal computer or workstation (16GB+ RAM recommended)	TensorFlow, TensorFlow Lite, OpenCV, Python visualization libraries	2 weeks	Project Proponents	N/A	Risk: Heatmap activation may not align precisely with the pathological region. Mitigation: Compare activation maps against veterinarian-annotated reference regions and refine visualization parameters.	Review heatmap localization quality and consistency against expert-annotated regions across all four classes.
First-Response Guidance and Output Synthesis Module	Develop the decision logic that synthesizes classification results, Grad-CAM output, and condition-matched first-response recommendations retrieved from a bundled, veterinarian-reviewed knowledge	Personal computer for development and testing	Flutter (Dart), bundled knowledge base, TensorFlow Lite	2 weeks	Project Proponents, Veterinary Professional (content review)	N/A	Risk: Mismatched or unclear guidance retrieval for a given classification. Mitigation: Validate condition-to-guidance mapping against the veterinarian-	Verify correct condition-to-guidance mapping and review clarity of first-response instructions for each class.

Module	Specific Objectives	Hardware Requirements	Software Requirements	Duration	Person Involved	Estimated Budget	Risk Assessment & Mitigation Strategies	Evaluation & Monitoring
	base into a single structured triage output						reviewed knowledge base and test all classification-guidance pairings.	
Offline Mobile Application and Image Processing Module	Develop the Flutter mobile application with offline functionality, on-device image quality filtering (blur and lighting checks), and the five-frame sequential capture and voting mechanism	Smartphone devices (Android/iOS), personal computer for development and testing	Flutter (Dart), TensorFlow Lite, tflite_flutter plugin, Android SDK	7 weeks	Project Proponents	N/A	Risk: Performance delays or inconsistent on-device inference across hardware variants. Mitigation: Optimize model size and quantization, and test across multiple device specifications.	Evaluate inference speed, offline availability, capture-to-result latency, and five-frame voting consistency.
System Testing and Evaluation Module	Conduct ISO/IEC 25010-based testing through white-box and black-box evaluation, including field evaluation with stray dog populations in Tacloban City, involving veterinary professionals and	Personal computer, smartphones for field testing	Automated testing tools, Google Forms/MS Forms, statistical analysis tools	6 weeks	Project Proponents, Veterinary Professionals, Community Responders	N/A	Risk: Inconsistent stakeholder feedback or field-testing constraints (e.g., limited stray dog accessibility). Mitigation: Standardize evaluation procedures,	Assess functional suitability, usability, reliability, classification accuracy, and stakeholder satisfaction using

Module	Specific Objectives	Hardware Requirements	Software Requirements	Duration	Person Involved	Estimated Budget	Risk Assessment & Mitigation Strategies	Evaluation & Monitoring
	community responders						schedule field sessions in coordination with local animal welfare partners, and conduct repeated validation sessions.	evaluation metrics and Likert scale results.

REFERENCES

Alanazi, J. M. (2025). Detection of Skin Lesions in Dogs using Advanced Convolutional Neural Network Technology. Indian Journal of Animal Research, Of. https://doi.org/10.18805/ijar.bf-1820

Ang, J., Tannaris, S., Chowanda, A., & Anderies, A. (2024). PetVision: Improved MobileViTv3-Based Image Classification for Cat Breed Identification. International Conference on Artificial Intelligence and Data Sciences (AiDAS), 452–456. https://doi.org/10.1109/aidas63860.2024.10730013

Attri, I., Vanita, B., Rajput, R., Awasthi, L. K., Thakur, A., Tripathi, D., Pathak, V., Shukla, P., & Gupta, D. (2025). SAAM-VetNet: an attention-based multi-task framework for animal disease detection and severity grading. Annals of Medicine and Surgery, 87(10), 6399–6408. https://doi.org/10.1097/ms9.0000000000003728

Badhon, S. M. S. I., Khushbu, S. A., Shaqib, S. M., Ali, M. A., Anik, A. H., & Hossain, K. S. M. T. (2025). Explainable AI for skin disease classification using gradient-weighted class activation mapping and transfer learning in digital health to identify contours. Digital Health, 11, 20552076251404523. https://doi.org/10.1177/20552076251404523

Baluran, J. M., et al. (2023). Pet-Care: An Android-based pet care management system for animal wellness. International Journal of Computing Sciences Research, 7, 2351-2368. https://stepacademic.net/ijcsr/article/view/364

Biju, S., Michael, S., Jose, T. M., Abraham, M., & Cherian, M. (2024). A review of machine learning approaches for canine skin disease detection using image processing techniques. https://ijera.in/index.php/IJERA/article/view/52

Campana, M. G., Colussi, M., Delmastro, F., Mascetti, S., & Pagani, E. (2023). A Transfer Learning and Explainable Solution to Detect mpox from Smartphones images. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2305.18489

Chinta, A., Thota, D. S., & Bypilla, A. D. (2024). Canine Demodicosis Detection in Dogs using Deep Learning. Conference on Intelligent Technologies (CONIT), 1–5. https://doi.org/10.1109/conit61985.2024.10626253

Coutinho, D. J. G., De Souza, J. S., Da Silva, J. M., & Teixeira, A. P. (2023). +CANINE ATOPIC DERMATITIS: DETAILED GUIDELINES FOR DIAGNOSIS AND ALLERGEN IDENTIFICATION. Revista Ibero-Americana De Humanidades, Ciências E Educação, 9(11), 71–88. https://doi.org/10.51891/rease.v9i11.12246

Hardi, S. M., Sutriyaningsih, & Ginting, D. S. B. (2025). Deep Learning for Veterinary Dermatology: A VGG16 Transfer Learning Approach for Skin Disease Classification in Pets. International Conference on Electrical, Telecommunication and Computer Engineering (ELTICOM), pp. 132-138, 132–138. https://doi.org/10.1109/elticom67568.2025.11336142

Harshitha, B. S., Pujitha, M. V., & Chandra, B. C. (2024). Canine Skin-Disease Detection Using CNN. International Conference on Knowledge Engineering and Communication Systems (ICKECS), 1–6. https://doi.org/10.1109/ickecs61492.2024.10616528

Hwang, S., Shin, H. K., Park, J. M., Kwon, B., & Kang, M. (2022). Classification of dog skin diseases using deep learning with images captured from multispectral imaging device. Molecular & Cellular Toxicology, 18(3), 299–309. https://doi.org/10.1007/s13273-022-00249-7

Idroes, G. M., Noviandy, T. R., Emran, T. B., & Idroes, R. (2024). Explainable Deep Learning Approach for Mpox Skin Lesion Detection with Grad-CAM. Heca Journal of Applied Sciences, 2(2), 54–63. https://doi.org/10.60084/hjas.v2i2.216

Inson, J., et al. (2021). Zoonotic potential of canine skin diseases: A survey of cases in Metro Manila, Philippines. Philippine Journal of Veterinary Medicine, 58(2), 112-120. https://pjvm.uplb.edu.ph/

Jalaboi, R., Faye, F., Orbes-Arteaga, M., Jørgensen, D., Winther, O., & Galimzianova, A. (2022). DermX: an end-to-end framework for explainable automated dermatological diagnosis. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2202.06956

Jia, L., Wang, T., Chen, Y., Zang, Y., Li, X., Shi, H., & Gao, L. (2023). MobileNet-CA-YOLO: an improved YOLOV7 based on the MobileNetV3 and Attention Mechanism for rice Pests and Diseases Detection. Agriculture, 13(7), 1285. https://doi.org/10.3390/agriculture13071285

Jose, T. M., Abraham, M., Biju, S., Michael, S., & Cherian, M. (2025). Canine Dermal Analyser: Harnessing artificial intelligence and deep learning to revolutionize canine skin disease detection. https://ijera.in/index.php/IJERA/article/view/236

Kamath, S., & Amrutha. (2025). A Deep Learning Approach for Detecting and Classifying Skin Conditions in Dogs from Dermatological Images. Zenodo (CERN European Organization for Nuclear Research). https://doi.org/10.5281/zenodo.15613205

Kaviti, V. S. (2025). Ensemble-Based automated skin disease classification using deep convolutional neural networks. International Journal for Research in Applied Science and Engineering Technology, 13(8), 413–420. https://doi.org/10.22214/ijraset.2025.73587

Kearley, M., DVM. (2025, February 28). Flea allergy Dermatitis in dogs: Everything a pet parent needs to know. PetMD. https://www.petmd.com/dog/conditions/skin/flea-allergy-dermatitis-dogs-everything-pet-parent-needs-know

Khaliq, S., Aziz, M. T., Safdar, B., Siddiq, U., Nawaz, F., & Khan, M. O. (2026). Zoonotic Parasites in Stray Dogs and Their Public Health Implications.Journal of Health Wellness and Community Research, 4(9), 1–10. https://doi.org/10.61919/r16cm079

Liu, D., & Suzuki, K. (2025). Edge- and Color–Texture-Aware Bag-of-Local-Features model for accurate and interpretable skin lesion diagnosis. Diagnostics, 15(15), 1883. https://doi.org/10.3390/diagnostics15151883

Madhavi, K., Kumar, C. V., Ranjith, M., & Hussein, L. (2025). Deep Learning Based Dog SkinPrediction: A Multi Label Classification Approach using Resent18. Third International Conference on Industry 4.0 Technology (I4Tech), 1–5. https://doi.org/10.1109/i4tech64670.2025.11277898

Mahendra, P. (2025). CURRENT UNDERSTANDING ON CANINE AND FELINE DERMATOPHYTOSIS: CLINICAL, DIAGNOSTIC AND THERAPEUTIC PERSPECTIVES. Journal of Bio Innovation, 297–304. https://doi.org/10.46344/jbino.2025.v14i02.10

Malawana, M.V.D.P.D., Suhail Razeeth, M.S., Alawathugoda, K.K.Y.M., and Kariapper, R.K.A.R., (2026). IoT and deep learning based dog dermatitis monitoring system. Journal of Information and Communication Technology (JICT), 3(2), 18–24.

Manoj, B.S., Prakruthi G., Vikas H.N. (2025). Evaluating Deep learning Model RESNET50 for dog skin Disease Classification: AI-Powered Dog Care Companion. International Research Journal of Innovations in Engineering and Technology, 09(Special Issue ICCIS), 86–91. https://doi.org/10.47001/irjiet/2025.iccis-202513

Martins, M., Kreutz, D., Basso, F., & Quincozes, S. E. (2024). X-Ray Covid19: Um Aplicativo Móvel para Diagnóstico através de Radiografias do Tórax. ESCOLA REGIONAL DE ENGENHARIA DE SOFTWARE (ERES), 79–88. https://doi.org/10.5753/eres.2024.4314

Moriello, K., Coyner, K., Paterson, S., & Mignon, B. (2025). Diagnosis and treatment of dermatophytosis in dogs and cats: Clinical Consensus Guidelines of the World Association for Veterinary Dermatology. Vet Dermatol. 2017;28(3):259-e68, 304-. https://www.merckvetmanual.com/integumentary-system/dermatophytosis/dermatophytosis-in-dogs-and-cats#For-More-Information_v49778424

Munjal, G., Bhardwaj, P., Bhargava, V., Singh, S., & Nagpal, N. (2024). SkinSage XAI: An explainable deep learning solution for skin lesion diagnosis. Health Care Science, 3(6), 438–455. https://doi.org/10.1002/hcs2.121

Nasir, S. (2025). DermAssist: A Hybrid Vision Transformer System for Skin Lesion Diagnosis with Automated Alerting and Dual-Sided Portals. medRxiv. https://doi.org/10.1101/2025.07.15.25331570

Natarajan, B., Nandhini, I., Syed, H.S., (2026). Enhancing Skin Lesion Diagnosis: Model Explainability Through Grad-CAM. International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI), 1–6. https://doi.org/10.1109/iciscois62701.2026.11447639

Newaz, A., Ishti, M. M., Azam, A. Z. M. A., & Adib, A. U. R. (2025). Toward Accessible Dermatology: Skin lesion Classification using Deep Learning Models on Mobile-Acquired Images. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2509.04800

Nguyen, P., Cao, D., Kha, Q. H., Chu, H., Le, M. H. N., Pham, T. Q. T., & Le, N. Q. K. (2026). JI-ADF: Joint-Individual Learning with Adaptive Decision Fusion for Multimodal Skin Lesion Classification. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2604.27343

Paryuni, A. D., Indarjulianto, S., & Widyarini, S. (2020). Dermatophytosis in companion animals: A review. Veterinary World, 13(6), 1174–1181. https://doi.org/10.14202/vetworld.2020.1174-1181

Patel, K., & Vishwakarma, V. (2025). Development of Deep Learning based Automated Detection and Classification of Dog Skin Diseases. International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI), 443–449. https://doi.org/10.1109/icoici65217.2025.11252820

Patel, K., & Vishwakarma, V. (2025). Development of Deep Learning based Automated Detection and Classification of Dog Skin Diseases. International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI), 443–449. https://doi.org/10.1109/icoici65217.2025.11252820

Racine, E., DVM. (2025, October 28). Hot spot treatments and prevention on dogs. American Kennel Club. https://www.akc.org/expert-advice/health/hot-spots-on-dogs/

Rathnayaka, R., Anuththara, K., Wickramasinghe, R., Gimhana, P., Weerasinghe, L., & Wimalaratne, G. (2022). Intelligent System for Skin Disease Detection of Dogs with Ontology Based Clinical Information Extraction. 2022 IEEE 13th Annual

Ubiquitous Computing, Electronics &Amp; Mobile Communication Conference (UEMCON), 0059–0066. https://doi.org/10.1109/uemcon54665.2022.9965696

Revathy, M., R, A., N, S., & R, S. (2025b). A MobileNet-Grad-Cam Framework for Visualizing and Interpreting Dermatological Image Classifications. International Conference on Computing Methodologies and Communication (ICCMC), 569–576, 569–576. https://doi.org/10.1109/iccmc65190.2025.11140625

Salces, M. J., et al. (2022). Prevalence and risk factors of zoonotic parasites in stray dogs in selected communities in the Philippines. Veterinary World, 15(5), 1245-1252. https://doi.org/10.14202/vetworld.2022.1245-1252

Shafiq, M., Saher, N., Alabdullah, B., Shahzad, F., Umer, M., Alnowaiser, K., Nam, Y., & Cho, Y. (2026). Feature fusion and explainable deep learning framework for intelligent skin disease classification using clinical dermatology images. International Journal of Computational Intelligence Systems, 19(1). https://doi.org/10.1007/s44196-025-01121-x

Sri, C. P., Nainika, D., Deepthi, J., & Devi, K. V. R. (2025). Dog Skin Problem Detection Using Deep Learning. Grenze International Journal of Engineering and Technology, 11(1).

Taranum, A., Metan, J., Yogegowda, P. A., & Krishnappa, C. D. (2024). Canine Disease Prediction using Multi-Directional Intensity Proportional Pattern with Correlated Textural Neural Network. The International Arab Journal of Information Technology, 21(5). https://doi.org/10.34028/iajit/21/5/11

Taufiqoh, S. D., & Purnamasari, P. D. (2025). Development of detection model for skin diseases in pets using image processing and deep learning techniques. International Journal of Electrical Computer and Biomedical Engineering, 3(2). https://doi.org/10.62146/ijecbe.v3i2.114

Thoutam, N., Ishika Mandloi, I., Kumari, A., Sonule, S., & Vrukshali Torawane, V. T. (2023). Detection and Classification of Dog Skin Disease using Deep Learning. International Journal of Scientific Research in Engineering and Management. https://ijsrem.com/download/detection-and-classification-of-dog-skin-disease-using-deep-learning

Upadhyay, A., Singh, G., Mhatre, S., & Nadar, P. (2023). Dog skin diseases detection and identification using convolutional neural networks. SN Computer Science, 4(3). https://doi.org/10.1007/s42979-022-01645-5

Yashu, Kukreja, V., Srivastava, P., Garg, A., & Hariharan, S. (2024). Paws and Pixels: Enhancing Canine Dermatology Through Image Analysis. Asia Pacific Conference on Innovation in Technology (APCIT), 1–6. https://doi.org/10.1109/apcit62007.2024.10673511

Zou, Y., Yang, G., Zhang, Q., Deng, S., & Tong, X. (2026). Multi-Spectral skin disease detection system based on Multi-Model Multi-Layer networks. Photonics, 13(2), 177. https://doi.org/10.3390/photonics13020177